

# US Patent & Trademark Office

## Patent Public Search | Text View

---

United States Patent Application Publication

20250285229

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Ikoma; Hayato

---

### MULTI-FOCUS IMAGE FUSION WITH BACKGROUND REMOVAL

---

#### Abstract

Provided herein are methods, systems, and computer program products for image fusion. A first z-stack of images of a biological sample may be received. The first z-stack of images may correspond to a first field of view, which comprises a plurality of patches. A second z-stack of images of the biological sample may be received. The second z-stack may correspond to the first field of view. A focus map may be determined based on the second z-stack, the focus map indicating, for each of a plurality of patches of the first field of view, one of the images of the second z-stack bringing into focus that patch of the first field of view. The focus map may be applied to the first and second z-stacks to generate respective first and second fused images. The first fused image may be subtracted from the second fused image to produce a subtracted image.

---

**Inventors:** Ikoma; Hayato (Dublin, CA)

**Applicant:** 10x Genomics, Inc. (Pleasanton, CA)

**Family ID:** 1000008504899

**Appl. No.:** 19/074119

**Filed:** March 07, 2025

#### Related U.S. Application Data

us-provisional-application US 63563095 20240308

---

#### Publication Classification

**Int. Cl.:** G06T5/50 (20060101); G06T3/4007 (20240101); G06T5/20 (20060101); G06T5/70 (20240101); G06T7/00 (20170101); G06T7/194 (20170101); G06T7/33 (20170101)

**U.S. Cl.:**

CPC      **G06T5/50** (20130101); **G06T3/4007** (20130101); **G06T5/20** (20130101); **G06T5/70** (20240101); **G06T7/194** (20170101); **G06T7/337** (20170101); **G06T7/97** (20170101); G06T2207/10024 (20130101); G06T2207/10148 (20130101); G06T2207/20032 (20130101); G06T2207/20221 (20130101); G06T2207/20224 (20130101); G06T2207/30024 (20130101)

---

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/563,095 filed Mar. 8, 2024, which is hereby incorporated by reference in its entirety.

### FIELD

[0002] The present disclosure is directed to systems and methods for multi-focus image fusion of a z-stack of images acquired from a sample (e.g., during in situ analysis). In particular, the present disclosure describes image processing of a z-stack of images obtained at different focal planes to generate a single, fused image that is focused across the entire image and also removing background (e.g., autofluorescence) from the resulting focused image.

### BACKGROUND

[0003] In situ detection and analysis methods are emerging from the rapidly developing field of spatial biology (e.g., spatial transcriptomics, spatial proteomics, etc.). The key objectives in spatial transcriptomics are to detect, quantify, and map gene activity to specific regions in a tissue sample at cellular or sub-cellular resolution. These techniques allow one to study the subcellular distribution of gene activity (as evidenced, e.g., by expressed gene transcripts), and have the potential to provide crucial insights in the fields of developmental biology, oncology, immunology, histology, etc.

[0004] For example, without being bound by theory or methodology, in situ detection and decoding are processes comprising a plurality of decoding cycles in each of which a different set of barcode probes (e.g., fluorescently-labeled oligonucleotides) is contacted with target analytes (e.g., mRNA sequences) or with target barcodes (e.g., nucleic acid barcodes) associated with the target analytes present in a sample (e.g., a tissue sample) under conditions that promote hybridization. One or more images (e.g., fluorescence images) are acquired in each decoding cycle, and the images are processed to detect the presence and locations of one or more barcode probes in each cycle. The presence and locations of one or more target analyte sequences or associated barcode sequences are then inferred from corresponding code words that are determined based on the set of, e.g., fluorescence signals detected in each decoding cycle of the decoding process.

[0005] When performing z-stack imaging of three-dimensional samples (e.g., a biological tissue), different features or portions of the same feature may be in focus at different focal planes. For example, when the feature being imaged is DAPI-stained nuclei, different nuclei may be positioned at different z-heights throughout the three-dimensional sample. Moreover, each feature (e.g., a DAPI-stained nucleus) extends in three-dimensions and, thus, will have a best-in-focus plane associated with the feature. When selecting an image from the z-stack for use in image processing, such as cell segmentation, a most in-focus plane may be selected from the z-stack. However, even if the most focused plane is selected from a z-stack of the biological tissue sample, it does not necessarily provide a focused image(s) of the biological tissue sample across the entire sample. Additionally, stained images, such as stained z-stack images that include immunofluorescence images captured by the imaging instrument, may be of a low contrast due to autofluorescence of the biological tissue sample. Therefore, there is an unmet need to generate a focused image(s) of the biological tissue sample across the entire sample. In addition, there is an unmet need to remove

the effects of the autofluorescence from the images to produce high contrast images.

## SUMMARY

[0006] Aspects of the claimed concepts provided herein provide methods, systems, and computer program products that produce focused and/or all-in-focus image(s) of a biological tissue sample, while also removing the effects of the autofluorescence from the immunofluorescence images to produce high contrast immunofluorescence images. The techniques, as described herein, are more energy and computationally efficient than existing techniques. Aspects of the claimed concepts as provided herein include a method of image fusion. A first z-stack of images of a biological sample is received. The first z-stack of images correspond to a first field of view, the first field of view comprising a plurality of patches. A second z-stack of images of the biological sample is received. The second z-stack of images corresponds to the first field of view. A focus map is determined based on the second z-stack. The focus map indicates, for each of a plurality of patches of the first field of view, one of the images of the second z-stack bringing into focus that patch of the first field of view. The focus map is applied to the first z-stack to generate a first fused image. The focus map is applied to the second z-stack to generate a second fused image. The first fused image is subtracted from the second fused image to produce a subtracted image.

[0007] The first z-stack of images may comprise background images of the biological sample, the second z-stack of images may comprise images of the biological sample stained with at least one stain, and the subtracted image may comprise a background-subtracted image. In various embodiments, the background images of the biological sample that may include nuclear staining, such as a DAPI stain. The first z-stack of images may comprise images of the biological sample that lacks cytoplasmic staining. The first z-stack of images may comprise images of the biological sample stained with a nuclear stain. The nuclear stain may comprise DAPI. The first z-stack of images may comprise images of the biological sample stained solely with DAPI. The second z-stack of images may comprise images of the biological sample stained with one or more fluorescent stain. The one or more fluorescent stains may comprise a nuclear stain. The one or more fluorescent stains may comprise one or more cytoplasmic stain. The one or more cytoplasmic stain may comprise at least one of: one or more ribosomal RNA stain, one or more lectin stain, and one or more antibody stain. The first z-stack of images and second z-stack of images may be acquired based on at least one probing cycle of the biological sample in an imaging instrument. The biological sample may comprise at least one cellular structure. The first z-stack of images and the second z-stack of images may be registered to each other based on the at least one cellular structure. The at least one cellular structure may comprise a nucleus. The first z-stack of images and the second z-stack of images each may comprise the nucleus. The registering may comprise registering the nucleus of the first z-stack to the nucleus of the second z-stack. The registering may comprise a scale-invariant feature transform (SIFT). The registering may comprise applying random sampling consensus (RANSAC).

[0008] Determining the focus map may include: determining a first focus metric for each image of the second z-stack; selecting one image of the second z-stack having a highest value of the first focus metric; selecting a set of adjacent images of the second z-stack including the selected image; determining a second focus metric for each of the plurality of patches of each of the selected set of images; and for each patch of the plurality of patches, selecting one of the selected set of images having a highest value of the second focus metric for that patch. The first focus metric may comprise Vollath's F4. The first focus metric may comprise a Tenengrad. The second focus metric may comprise a Tenengrad. The second focus metric may comprise Vollath's F4. Each patch in the plurality of patches may comprise a same shape. The shape may comprise a square shape. The shape may be about 16×16 pixels to about 128×128 pixels in size. Each selected image of the selected set of images may have an associated index indicating a position within the second z-stack of images. The focus map may indicate the index associated with the selected image for each patch. One or more patch having more than a predetermined disparity between its associated index and

the indices associated with two or more neighboring patches may be identified. The index associated with the identified one or more patch may be replaced with an interpolated index based on the indices of the two or more neighboring patches. The predetermined disparity may be 2, 3, 4, 5, 6, 7, 8, 9, or 10. A median filter may be applied to the indices of the focus map. The focus map may be upsampled to pixel resolution. The upsampling may comprise linear interpolation. The set of adjacent images may be of a predetermined size. The predetermined size may be 2 images to 31 images. The predetermined size may be 21 images. A denoising filter may be applied to each image of the selected set of images. Applying the focus map to the first z-stack of images may comprise: sampling the first z-stack of images according to the focus map to generate a first intermediate image; generating a shifted focus map from the focus map; sampling the first z-stack of images according to the shifted focus map to generate a second intermediate image; and determining a convex combination of the first intermediate image and the second intermediate image to generate the first fused image. Determining the convex combination may comprise a random sampling consensus. The subtracted image may be stitched together with one or more additional subtracted images based on the field of view and a color channel of the subtracted image.

[0009] A computer program product may include a computer readable storage medium having program instructions embodied therewith, the program instructions may be executable by a processor to cause the processor to perform the method as described herein.

[0010] A system may include an imaging instrument; and a computing node operatively coupled to the imaging instrument comprising a computer readable storage medium having program instructions embodied therewith, the program instructions being executable by a processor to cause the processor to perform the method as described herein.

[0011] Aspects of the claimed concepts as provided herein include a method of image fusion. A z-stack of images of a biological sample is received. The z-stack of images correspond to a field of view. A focus map is determined based on the z-stack of images, wherein determining the focus map comprises: determining a focus metric for each image of the z-stack of images; selecting one image of the z-stack having a highest value of the focus metric; selecting a set of adjacent images of the z-stack including the selected image; dividing each image in the selected set of images into a plurality of patches; determining a second focus metric for each of the plurality of patches of each of the selected set of images; and for each patch of the plurality of patches, selecting one of the selected set of images having a highest value of the second focus metric for that patch. The focus map is applied to the z-stack to generate a fused image.

[0012] The z-stack of images may comprise images of the biological sample stained with one or more fluorescent stain. The one or more fluorescent stains may comprise a nuclear stain. The one or more fluorescent stains may comprise one or more cytoplasmic stain. The one or more cytoplasmic stain may comprise at least one of: one or more ribosomal RNA stain, one or more lectin stain, and one or more antibody stain. The first focus metric may comprise Vollath's F4. The second focus metric may comprise Tenengrad. The first focus metric may comprise Tenengrad. The second focus metric may comprise Vollath's F4. Each patch in the plurality of patches may comprise a same shape. The shape may comprise a square shape. The shape may be about 16×16 pixels to about 128×128 pixels in size. Each selected image of the set of selected images may have an associated index indicating a position within the z-stack of images. The focus map may indicate the index associated with the selected image for each patch. One or more patch having more than a predetermined disparity between its associated index and the indices associated with two or more neighboring patches may be identified. The index associated with the identified one or more patch may be replaced with an interpolated index based on the indices of the two or more neighboring patches. The predetermined disparity may be 2, 3, 4, 5, 6, 7, 8, 9, or 10. A median filter may be applied to the indices of the focus map. The focus map may be upsampled to pixel resolution. The upsampling may comprise linear interpolation. The set of adjacent images may be of a

predetermined size. The predetermined size may be 2 images to 31 images. The predetermined size may be 21 images. A denoising filter may be applied to each image of the selected set of images.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the office upon request and payment of the necessary fee.

[0014] FIG. 1 depicts an overview of a volumetric sample imaging system and illustrates a Field of View (FOV) grid bounding the sample (e.g., hydrogel, tissue section, one or more cells, etc.) as projected onto the surface of a solid substrate supporting the sample.

[0015] FIG. 2 depicts the XZ cross-sectional view and illustrates the plurality of images in the Z dimension. The objective lens focal point is positioned to acquire an image at every Z-slice in a Z-stack. An XZ image of signal distribution within the imaging volume is shown (bottom).

[0016] FIG. 3 is an example workflow of image data acquisition from a biological sample (e.g., a cell or tissue sample) using an opto-fluidic instrument, according to various embodiments.

[0017] FIGS. 4A-4B illustrate cross-sectional views of an optics module in an imaging system, according to some embodiments.

[0018] FIG. 5 depicts a computing node according to some embodiments disclosed herein.

[0019] FIG. 6 illustrates a z-slice image exhibiting distortion at the edges of the FOV due to field curvature, according to some embodiments.

[0020] FIG. 7 illustrates various cycles of an optofluidic instrument used to generate a final stain image using multi-focus image fusion, according to some embodiments.

[0021] FIG. 8 depicts a visualization of a z-slice of a multi-channel z-stack of images of a biological sample (e.g., a background image with only DAPI staining), according to various embodiments.

[0022] FIG. 9 depicts a visualization of a z-slice of a multi-channel z-stack of images of a stained biological sample, according to various embodiments.

[0023] FIGS. 10A-10H depict aspects of steps of a multi-focus image fusion algorithm, according to various embodiments.

[0024] FIG. 11 depicts a technique for determining a subtracted image by subtracting a convex combination of fused unstained images from the stained fused image, according to various embodiments.

[0025] FIG. 12 depicts a visualization of a finalized subtracted focused image, according to various embodiments.

[0026] FIG. 13 is a flowchart illustrating a method of image fusion, according to embodiments of the present disclosure.

[0027] FIG. 14 is a flowchart illustrating a method of image fusion, according to embodiments of the present disclosure.

[0028] In the figures, elements and steps having the same or similar reference numeral have the same or similar attributes or description, unless explicitly stated otherwise.

### DETAILED DESCRIPTION

[0029] In volumetric sample imaging systems (e.g., an optofluidic instrument), a z-stack of images is obtained for each Field of View (FOV) of the objective (see FIG. 1), wherein the imaged regions contain target molecules, such as nucleic acids or proteins, and relevant feature data (e.g., morphologic and pathologic data) (see FIG. 2). The resulting datasets become exceedingly large, comprising millions of data points and, therefore, become increasingly challenging to visualize for analysis.

[0030] Provided herein are approaches for representing features of in situ data (e.g., sequencing data, morphological data, and the like, typically acquired using microscopy-based methods from biological samples) that is scalable for up to any number of feature positions, by subsampling based on the current zoom level. Such an approach can be used to express hundreds of millions of points (whereas alternative approaches to statically load and render points is limited to 5 to 10 million points before it becomes unusable). The in-situ visualization application disclosed herein provides mechanisms that allow the end user to interactively explore data overlays, representing quantitative and qualitative information, atop the images representing the morphological structure of the underlying tissue sample.

[0031] For example, and without limitation, the in situ visualization tools disclosed herein may rely on cloud-hosted data storage, a server application that responds to web browser requests, and a client application that operates entirely within a web browser. In situ data may be accessed through an interactive visual interface, e.g., a standalone web browser, a desktop application, or an on-instrument interface.

[0032] The client application may first initiate a request to a running instance of the visualization server, by either entering a web address or clicking a link within a browser, or following a workflow within an instrument interface. The visualization server may then provide the browser with the client-side code to load. Depending on the parameters of the initial request, the server response may include settings or other information derived from the parameters, such as identifying information and the location of the desired dataset, the version of the client-side code to load, and other metadata. In some embodiments, the request is initiated to an application running on an instrument, or on a dedicated machine within an end-user's on-premise computing infrastructure.

[0033] Once the client-side code is loaded, the client may make requests to the visualization server for image resources, which may primarily be used to display microscopy-based morphological data. In some embodiments, images may be stored in the cloud, or locally on a high-capacity computer, or within a shared networked filesystem. Said image resources may comprise multiple sets of images, each corresponding to a different wavelength, time interval, z-slice, or iteration of the in situ instrument's imaging cycle. Without being bound by theory and for exemplary purposes, these images can capture the presence or absence of proteins that can label cell boundaries, nuclear and cytoplasmic boundaries, the presence or absence of targeted proteins, or other morphological or fine-grained molecular information. In some embodiments, the image data can be 8-bit or 16-bit single channel, or 8 bpp/16 bpp multichannel RGBA images. In some embodiments, images can be stored natively in TIFF or other image formats.

[0034] In addition to underlying image data, the client application may make requests for the files that will drive data overlays atop the morphological data. For example and without limitation, such data may include quantitative per-cell information, such as the number of molecules per gene per cell, cluster membership information, per-cell protein or other gene marker statistics, cell type identification, morphological tissue boundaries, and more. The server and client application transform the data from its at-rest format in storage to an in-memory format suitable for the visualization tools, such as arrays of colors, polygons, or shading parameters. The visualization user interface may then be used to control, for example, which overlays are active, tuning of said overlays, and allow a user to more easily explore and characterize phenomena found in the in situ data.

[0035] Described herein is a spatial transcriptomics solution and technique, which allow for the analysis of the transcriptome within a tissue sample.

[0036] The spatial transcriptomics solution, as described herein may include an in situ capturing technique. This may allow for the transcript to be captured within a tissue sample. After capture of the transcript, sequencing may be performed outside the tissue sample. Such an approach may allow for the tracing back of the transcripts, such as RNA transcripts, to their original location in the tissue sample. The tissue sample may be a particular type of tissue sample such as a fresh-

frozen tissue sample or a Formalin Fixed Paraffin Embedded (FFPE) tissue sample.

[0037] The spatial transcriptomics solution may allow for a determination of where in a tissue sample a particular gene is expressed. The spatial transcriptomics solution may allow for a determination of where a cluster of genes that are expressed are located.

[0038] The spatial transcriptomics solution may capture data using slides, which each include one or more (e.g., four) capture areas. Each capture area on a slide may include multiple barcoded spots (e.g., 5000 barcoded spots). Each barcoded spot may include capture oligonucleotides. Each such capture oligonucleotide may bind to RNA in the tissue sample that is applied to the slide. Each barcoded spot may capture the transcripts from a particular number (e.g., 1-10) cells within the tissue sample.

[0039] A tissue sample may be applied (e.g., placed onto) a slide so that information may be captured from the tissue sample via the capture areas of the slide. The tissue sample may be permeabilized. This may allow the tissue sample to release mRNA, to which the capture oligonucleotides, associated with the spots in the slide capture areas, may bind. In addition, the tissue sample may be stained using IF or H&E stains. Image(s) of the stained tissue sample may be taken. The captured mRNA may be synthesized into cDNA. Sequencing libraries may be prepared therefrom. The transcriptomics data may contain a spatial barcode from the spot on the slide. This may allow for the linking of the transcriptomics data to the location on the slide (e.g., regions of an image of the biological sample on the slide), as further described herein.

#### General Terminology

[0040] Specific terminology is used throughout this disclosure to explain various aspects of the methods, systems, and compositions that are described. Unless otherwise defined, all of the technical terms used herein have the same meaning as commonly understood by one of ordinary skill in the art in the field to which this disclosure belongs.

[0041] As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. For example, “a” or “an” means “at least one” or “one or more”. Any reference to “or” herein is intended to encompass “and/or” unless otherwise stated.

[0042] As used herein, the terms “comprising” (and any form or variant of comprising, such as “comprise” and “comprises”), “having” (and any form or variant of having, such as “have” and “has”), “including” (and any form or variant of including, such as “includes” and “include”), or “containing” (and any form or variant of containing, such as “contains” and “contain”), are inclusive or open-ended and do not exclude additional, un-recited additives, components, integers, elements or method steps.

[0043] As used herein, the term “about” a number refers to that number plus or minus 10% of that number. The term ‘about’ when used in the context of a range refers to that range minus 10% of its lowest value and plus 10% of its greatest value.

[0044] Throughout this disclosure, various aspects of the claimed subject matter are presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the claimed subject matter. Accordingly, the description of a range should be considered to have specifically disclosed all the possible sub-ranges as well as individual numerical values within that range. For example, where a range of values is provided, it is understood that each intervening value, between the upper and lower limit of that range and any other stated or intervening value in that stated range is encompassed within the claimed subject matter. The upper and lower limits of these smaller ranges may independently be included in the smaller ranges, and are also encompassed within the claimed subject matter, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the claimed subject matter. This applies regardless of the breadth of the range.

[0045] Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements. Similarly, use of a), b), etc., or i), ii), etc. does not by itself connote any priority, precedence, or order of steps in the claims. Similarly, the use of these terms in the specification does not by itself connote any required priority, precedence, or order.

[0046] The term “platform” (or “system”) may refer to an ensemble of: (i) instruments (e.g., imaging instruments, fluid controllers, temperature controllers, motion controllers and translation stages, etc.), (ii) devices (e.g., specimen slides, substrates, flow cells, microfluidic devices, etc., which may comprise fixed and/or removable or disposable components of the platform), (iii) reagents and/or reagent kits, and (iv) software, or any combination thereof, which allows a user to perform one or more bioassay methods (e.g., analyte detection, in situ detection or sequencing, and/or nucleic acid detection or sequencing) depending on the particular combination of instruments, devices, reagents, reagent kits, and/or software utilized. As used herein, the term sequencing may include sequencing by synthesis (SBS), sequencing by hybridization (SBH), sequencing by ligation (SBL), sequencing by binding (SBB), and/or any other type of sequencing. SBS may be a DNA sequencing technique in which fluorescently labeled nucleotides are used to sequence clusters on a flow cell surface. SBH may be a DNA sequencing technique in which sets of oligonucleotides are hybridized under conditions that allow detection of complementary sequences in the target nucleic acid. SBL may be a DNA sequencing technique that uses the enzyme DNA ligase to identify the nucleotide present at a given position in a DNA sequence. SBB may be a DNA sequencing technique that involves the examination of a ternary complex that forms between a primer-template nucleic acid hybrid, polymerase and nucleotide triphosphate and acquiring signal that is used to determine nucleic acid base identity without the need for nucleotide incorporation.

[0047] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

#### Barcoding and Decoding Terminology

[0048] A “barcode” is a label, or identifier, that conveys or is capable of conveying information (e.g., information about an analyte in a sample, a cell, a bead, a location, a sample, and/or a capture probe). The term “barcode” may refer either to a physical barcode molecule (e.g., a nucleic acid barcode molecule) or to its representation in a computer-readable, digital format (e.g., as a string of characters representing the sequence of bases in a nucleic acid barcode molecule).

[0049] The phrase “barcode diversity” refers to the total number of unique barcode sequences that may be represented by a given set of barcodes.

[0050] A physical barcode molecule (e.g., a nucleic acid barcode molecule) that forms a label or identifier as described above. In some instances, a barcode can be part of an analyte, can be independent of an analyte, can be attached to an analyte, or can be attached to or part of a probe that targets the analyte. In some instances, a particular barcode can be unique relative to other barcodes.

[0051] Physical barcodes can have a variety of different formats. For example, barcodes can include polynucleotide barcodes, random nucleic acid and/or amino acid sequences, and synthetic nucleic acid and/or amino acid sequences. A physical barcode can be attached to an analyte, or to another moiety or structure, in a reversible or irreversible manner. A physical barcode can be added to, for example, a fragment of a deoxyribonucleic acid (DNA) or ribonucleic acid (RNA) sample before or during sequencing of the sample. In some instances, barcodes can allow for identification and/or quantification of individual sequencing-reads in sequencing-based methods (e.g., a barcode can be or can include a unique molecular identifier or “UMI”). Barcodes can be used to detect and spatially-resolve molecular components found in biological samples, for example, at single-cell



resolution (e.g., a barcode can be, or can include, a molecular barcode, a spatial barcode, a unique molecular identifier (UMI), etc.).

[0052] In some instances, barcodes may comprise a series of two or more segments or sub-barcodes (e.g., corresponding to “letters” or “code words” in a decoded barcode), each of which may comprise one or more of the subunits or building blocks used to synthesize the physical (e.g., nucleic acid) barcode molecules. For example, a nucleic acid barcode molecule may comprise two or more barcode segments, each of which comprises one or more nucleotides. In some instances, a barcode may comprise at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more than 10 segments. In some instances, each segment of a barcode molecule may comprise at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20, or more than 20 subunits or building blocks. For example, each segment of a nucleic acid barcode molecule may comprise at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20, or more than 20 nucleotides. In some instances, two or more of the segments of a barcode may be separated by non-barcode segments, i.e., the segments of a barcode molecule need not be contiguous.

[0053] A “digital barcode” (or “digital barcode sequence”) is a representation of a corresponding physical barcode (or target analyte sequence) in a computer-readable, digital format as described above. A digital barcode may comprise one or more “letters” (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20, or more than 20 letters) or one or more “code words” (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more than 10 code words), where a “code word” comprises, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20, or more than 20 letters. In some instances, the sequence of letters or code words in a digital barcode sequence may correspond directly with the sequence of building blocks (e.g., nucleotides) in a physical barcode. In some instances, the sequence of letters or code words in a digital barcode sequence may not correspond directly with the sequence of building blocks in a physical barcode, but rather may comprise, e.g., arbitrary code words that each correspond to a segment of a physical barcode. For example, in some instances, the disclosed methods for decoding and error correction may be applied directly to detecting target analyte sequences (e.g., mRNA sequences) as opposed to detecting target barcodes, and the barcode probes used to detect the target analyte sequences may correspond to letters or code words that have been assigned to specific target analyte sequences but that do not directly correspond to the target analyte sequences.

[0054] A “designed barcode” (or “designed barcode sequence”) is a barcode (or its digital equivalent; in some instances a designed barcode may comprise a series of code words that can be assigned to gene transcripts and subsequently decoded into a decoded barcode) that meets a specified set of design criteria as required for a specific application. In some instances, a set of designed barcodes may comprise at least 2, at least 5, at least 10, at least 20, at least 40, at least 60, at least 80, at least 100, at least 200, at least 400, at least 600, at least 800, at least 1,000, at least 2,000, at least 4,000, at least 6,000, at least 8,000, at least 10,000, at least 20,000, at least 40,000, at least 60,000, at least 80,000, at least 100,000, at least 200,000, at least 400,000, at least 600,000, at least 800,000, at least 1,000,000, at least  $2 \times 10^6$ , at least  $3 \times 10^6$ , at least  $4 \times 10^6$ , at least  $5 \times 10^6$ , at least  $6 \times 10^6$ , at least  $7 \times 10^6$ , at least  $8 \times 10^6$ , at least  $9 \times 10^6$ , at least  $10^7$ , at least  $10^8$ , at least  $10^9$ , or more than  $10^9$  unique barcodes. In some instances, a set of designed barcodes may comprise any number of designed barcodes within the range of values in this paragraph, e.g., 1,225 unique barcodes or  $2.38 \times 10^6$  unique barcodes. As noted above for barcodes in general, in some instances designed barcodes may comprise two or more segments (corresponding to two or more code words in a decoded barcode). In those cases, the specified set of design criteria may be applied to the designed barcodes as a whole, or to one or more segments (or positions) within the designed barcodes.

[0055] A “decoded barcode” (or “decoded barcode sequence”) is a digital barcode sequence generated via a decoding process that ideally matches a designed barcode sequence, but that may include errors arising from noise in the synthesis process used to create barcodes and/or noise in the decoding process itself. As noted above, in some instances, the disclosed methods for decoding and error correction may be applied directly to detecting target analytes (e.g., mRNA sequences) as

opposed to detecting target barcodes, and the barcode probes used to detect the target analytes may correspond to letters or code words that have been assigned to specific target analytes but that do not directly correspond to the target analytes. In these instances, a decoded barcode (i.e., a series of letters or code words) may serve as a proxy for the target analyte.

[0056] A “corrected barcode” (or “corrected barcode sequence”) is a digital barcode sequence derived from a decoded barcode sequence by applying one or more error correction methods.

#### Probe Terminology

[0057] The term “probe” may refer either to a physical probe molecule (e.g., a nucleic acid probe molecule) or to its representation in a computer-readable, digital format (e.g., as a string of characters representing the sequence of bases in a nucleic acid probe molecule). A “probe” may be, for example, a molecule designed to recognize (and bind or hybridize to) another molecule, e.g., a target analyte, another probe molecule, etc.

[0058] In some instances, a physical probe molecule may comprise one or more of the following: (i) a target recognition element (e.g., an antibody capable of recognizing and binding to a target peptide, protein, or small molecule; an oligonucleotide sequence that is complementary to a target gene sequence or gene transcript; or a poly-T oligonucleotide sequence that is complementary to the poly-A tails on messenger RNA molecules), (ii) a barcode element (e.g., a molecular barcode, a cell barcode, a spatial barcode, and/or a unique molecular identifier (UMI)), (iii) an amplification and/or sequencing primer binding site, (iv) one or more linker regions, (v) one or more detectable tags (e.g., fluorophores), or any combination thereof. In some instances, each component of a probe molecule may comprise at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20, or more than 20 subunits or building blocks. For example, in some instances, each component of a nucleic acid probe molecule may comprise at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20, or more than 20 nucleotides.

[0059] In some instances, physical probes may bind or hybridize directly to their target. In some instances, physical probes may bind or hybridize indirectly to their target. For example, in some instances, a secondary probe may bind or hybridize to a primary probe, where the primary probe binds or hybridizes directly to the target analyte. In some instances, a tertiary probe may bind or hybridize to a secondary probe, where the secondary probe binds or hybridizes to a primary probe, and where the primary probe binds or hybridizes directly to the target analyte.

[0060] Examples of “probes” and their applications include, but are not limited to, primary probes (e.g., molecules designed to recognize and bind or hybridize to target analyte), intermediate probes (e.g., molecules designed to recognize and bind or hybridize to another molecule and provide a hybridization or binding site for another probe (e.g., a detection probe), detection probes (e.g., molecules designed to recognize and bind or hybridize to another molecule, detection probes may be labeled with a fluorophore or other detectable tag). In some instances, a probe may be designed to recognize and bind (or hybridize) to a physical barcode sequence (or segments thereof). In some instances, a probe may be used to detect and decode a barcode, e.g., a nucleic acid barcode. In some instances, a probe may bind or hybridize directly to a target barcode. In some instances, a probe may bind or hybridize indirectly to a target barcode (e.g., by binding or hybridizing to other probe molecules which itself is bound or hybridized to the target barcode).

#### Nucleic Acid Molecule and Nucleotide Terminology

[0061] The terms “nucleic acid” (or “nucleic acid molecule”) and “nucleotide” are intended to be consistent with their use in the art and to include naturally-occurring species or functional analogs thereof. Particularly useful functional analogs of nucleic acids are capable of hybridizing to a nucleic acid in a sequence-specific fashion (e.g., capable of hybridizing to two nucleic acids such that ligation can occur between the two hybridized nucleic acids) or are capable of being used as a template for replication of a particular nucleotide sequence. Naturally-occurring nucleic acids generally have a backbone containing phosphodiester bonds. An analog structure can have an alternate backbone linkage including any of a variety of those known in the art. Naturally-occurring

nucleic acids generally have a deoxyribose sugar (e.g., found in deoxyribonucleic acid (DNA)) or a ribose sugar (e.g. found in ribonucleic acid (RNA)).

[0062] A nucleic acid can contain nucleotides having any of a variety of analogs of these sugar moieties that are known in the art. A nucleic acid can include natural or non-natural nucleotides. In this regard, a naturally-occurring deoxyribonucleic acid can have one or more bases selected from the group consisting of adenine (A), thymine (T), cytosine (C), or guanine (G), and a ribonucleic acid can have one or more bases selected from the group consisting of uracil (U), adenine (A), cytosine (C), or guanine (G). Useful non-natural bases that can be included in a nucleic acid or nucleotide are known in the art. See, for example, Appella (2009), “Non-Natural Nucleic Acids for Synthetic Biology”, *Curr Opin Chem Biol.* 13(5-6): 687-696; and Duffy, et al. (2020), “Modified Nucleic Acids: Replication, Evolution, and Next-Generation Therapeutics”, *BMC Biology* 18:112. Samples:

[0063] A sample disclosed herein can be or derived from any biological sample. Methods and compositions disclosed herein may be used for analyzing a biological sample, which may be obtained from a subject using any of a variety of techniques including, but not limited to, biopsy, surgery, and laser capture microscopy (LCM), and generally includes cells and/or other biological material from the subject. In addition to the subjects described above, a biological sample can be obtained from a prokaryote such as a bacterium, an archaea, a virus, or a viroid. A biological sample can also be obtained from non-mammalian organisms (e.g., a plant, an insect, an arachnid, a nematode, a fungus, or an amphibian). A biological sample can also be obtained from a eukaryote, such as a tissue sample, a patient derived organoid (PDO) or patient derived xenograft (PDX). A biological sample from an organism may comprise one or more other organisms or components therefrom. For example, a mammalian tissue section may comprise a prion, a viroid, a virus, a bacterium, a fungus, or components from other organisms, in addition to mammalian cells and non-cellular tissue components. Subjects from which biological samples can be obtained can be healthy or asymptomatic individuals, individuals that have or are suspected of having a disease (e.g., a patient with a disease such as cancer) or a pre-disposition to a disease, and/or individuals in need of therapy or suspected of needing therapy.

[0064] The biological sample can include any number of macromolecules, for example, cellular macromolecules and organelles (e.g., mitochondria and nuclei). The biological sample can be a nucleic acid sample and/or protein sample. The biological sample can be a carbohydrate sample or a lipid sample. The biological sample can be obtained as a tissue sample, such as a tissue section, biopsy, a core biopsy, needle aspirate, or fine needle aspirate. The sample can be a fluid sample, such as a blood sample, urine sample, or saliva sample. The sample can be a skin sample, a colon sample, a cheek swab, a histology sample, a histopathology sample, a plasma or serum sample, a tumor sample, living cells, cultured cells, a clinical sample such as, for example, whole blood or blood-derived products, blood cells, or cultured tissues or cells, including cell suspensions. In some instances, the biological sample may comprise cells which are deposited on a surface.

[0065] Cell-free biological samples can include extracellular macromolecules, e.g., polynucleotides. Extracellular polynucleotides can be isolated from a bodily sample, e.g., blood, plasma, serum, urine, saliva, mucosal excretions, sputum, stool, and tears.

[0066] Biological samples can be derived from a homogeneous culture or population of the subjects or organisms mentioned herein or alternatively from a collection of several different organisms, for example, in a community or ecosystem.

[0067] Biological samples can include one or more diseased cells. A diseased cell can have altered metabolic properties, gene expression, protein expression, and/or morphologic features. Examples of diseases include inflammatory disorders, metabolic disorders, nervous system disorders, and cancer. Cancer cells can be derived from solid tumors, hematological malignancies, cell lines, or obtained as circulating tumor cells. Biological samples can also include fetal cells and immune cells.

[0068] In some instances, a substrate herein can be any support that is insoluble in aqueous liquid and which allows for positioning of biological samples, analytes, features, and/or reagents (e.g., probes) on the support. In some instances, a biological sample can be attached to a substrate. Attachment of the biological sample can be irreversible or reversible, depending upon the nature of the sample and subsequent steps in the analytical method. In certain instances, the sample can be attached to the substrate reversibly by applying a suitable polymer coating to the substrate, and contacting the sample to the polymer coating. The sample can then be detached from the substrate, e.g., using an organic solvent that at least partially dissolves the polymer coating. In some instances, the substrate can be coated or functionalized with one or more substances to facilitate attachment of the sample to the substrate. Suitable substances that can be used to coat or functionalize the substrate include, but are not limited to, lectins, poly-lysine, antibodies, and polysaccharides.

[0069] A variety of steps can be performed to prepare or process a biological sample for and/or during an assay. Except where indicated otherwise, the preparative or processing steps described below can generally be combined in any manner and in any order to appropriately prepare or process a particular sample for and/or analysis.

#### Endogenous Analytes:

[0070] In some instances, an analyte herein is endogenous to a biological sample and can include nucleic acid analytes and non-nucleic acid analytes. Methods and compositions disclosed herein can be used to analyze nucleic acid analytes (e.g., using a nucleic acid probe or probe set that directly or indirectly hybridizes to a nucleic acid analyte) and/or non-nucleic acid analytes (e.g., using a labelling agent that comprises a reporter oligonucleotide and binds directly or indirectly to a non-nucleic acid analyte) in any suitable combination.

[0071] Examples of non-nucleic acid analytes include, but are not limited to, lipids, carbohydrates, peptides, proteins, glycoproteins (N-linked or O-linked), lipoproteins, phosphoproteins, specific phosphorylated or acetylated variants of proteins, amidation variants of proteins, hydroxylation variants of proteins, methylation variants of proteins, ubiquitylation variants of proteins, sulfation variants of proteins, viral coat proteins, extracellular and intracellular proteins, antibodies, and antigen binding fragments. In some instances, the analyte is inside a cell or on a cell surface, such as a transmembrane analyte or one that is attached to the cell membrane. In some instances, the analyte can be an organelle (e.g., nuclei or mitochondria). In some instances, the analyte is an extracellular analyte, such as a secreted analyte. Exemplary analytes include, but are not limited to, a receptor, an antigen, a surface protein, a transmembrane protein, a cluster of differentiation protein, a protein channel, a protein pump, a carrier protein, a phospholipid, a glycoprotein, a glycolipid, a cell-cell interaction protein complex, an antigen-presenting complex, a major histocompatibility complex, an engineered T-cell receptor, a T-cell receptor, a B-cell receptor, a chimeric antigen receptor, an extracellular matrix protein, a posttranslational modification (e.g., phosphorylation, glycosylation, ubiquitination, nitrosylation, methylation, acetylation or lipidation) state of a cell surface protein, a gap junction, and an adherens junction.

[0072] Examples of nucleic acid analytes include DNA analytes such as single-stranded DNA (ssDNA), double-stranded DNA (dsDNA), genomic DNA, methylated DNA, specific methylated DNA sequences, fragmented DNA, mitochondrial DNA, in situ synthesized PCR products, and RNA/DNA hybrids. The DNA analyte can be a transcript of another nucleic acid molecule (e.g., DNA or RNA such as mRNA) present in a tissue sample.

[0073] Examples of nucleic acid analytes also include RNA analytes such as various types of coding and non-coding RNA. Examples of the different types of RNA analytes include messenger RNA (mRNA), including a nascent RNA, a pre-mRNA, a primary-transcript RNA, and a processed RNA, such as a capped mRNA (e.g., with a 5' 7-methyl guanosine cap), a polyadenylated mRNA (poly-A tail at 3' end), and a spliced mRNA in which one or more introns have been removed. Also included in the analytes disclosed herein are non-capped mRNA, a non-polyadenylated mRNA, and

a non-spliced mRNA. The RNA analyte can be a transcript of another nucleic acid molecule (e.g., DNA or RNA such as viral RNA) present in a tissue sample. Examples of a non-coding RNAs (ncRNA) that is not translated into a protein include transfer RNAs (tRNAs) and ribosomal RNAs (rRNAs), as well as small non-coding RNAs such as microRNA (miRNA), small interfering RNA (siRNA), Piwi-interacting RNA (piRNA), small nucleolar RNA (snoRNA), small nuclear RNA (snRNA), extracellular RNA (exRNA), small Cajal body-specific RNAs (scaRNAs), and the long ncRNAs such as Xist and HOTAIR. The RNA can be small (e.g., less than 200 nucleic acid bases in length) or large (e.g., RNA greater than 200 nucleic acid bases in length). Examples of small RNAs include 5.8S ribosomal RNA (rRNA), 5S rRNA, tRNA, miRNA, siRNA, snoRNAs, piRNA, tRNA-derived small RNA (tsRNA), and small rDNA-derived RNA (srRNA). The RNA can be double-stranded RNA or single-stranded RNA. The RNA can be circular RNA. The RNA can be a bacterial rRNA (e.g., 16s rRNA or 23s rRNA).

[0074] In some instances described herein, an analyte may be a denatured nucleic acid, wherein the resulting denatured nucleic acid is single-stranded. The nucleic acid may be denatured, for example, optionally using formamide, heat, or both formamide and heat. In some instances, the nucleic acid is not denatured for use in a method disclosed herein.

[0075] In certain instances, an analyte can be extracted from a live cell. Processing conditions can be adjusted to ensure that a biological sample remains live during analysis, and analytes are extracted from (or released from) live cells of the sample. Live cell-derived analytes can be obtained only once from the sample, or can be obtained at intervals from a sample that continues to remain in viable condition.

[0076] Methods and compositions disclosed herein can be used to analyze any number of analytes. For example, the number of analytes that are analyzed can be at least about 2, at least about 3, at least about 4, at least about 5, at least about 6, at least about 7, at least about 8, at least about 9, at least about 10, at least about 11, at least about 12, at least about 13, at least about 14, at least about 15, at least about 20, at least about 25, at least about 30, at least about 40, at least about 50, at least about 100, at least about 1,000, at least about 10,000, at least about 100,000 or more different analytes present in a region of the sample or within an individual feature of the substrate.

[0077] In any implementation described herein, the analyte comprises a target sequence. In some instances, the target sequence may be endogenous to the sample, generated in the sample, added to the sample, or associated with an analyte in the sample. In some instances, the target sequence is a single-stranded target sequence (e.g., a sequence in a rolling circle amplification product). In some instances, the analytes comprise one or more single-stranded target sequences. In one aspect, a first single-stranded target sequence is not identical to a second single-stranded target sequence. In another aspect, a first single-stranded target sequence is identical to one or more second single-stranded target sequence. In some instances, the one or more second single-stranded target sequence is comprised in the same analyte (e.g., nucleic acid) as the first single-stranded target sequence. Alternatively, the one or more second single-stranded target sequence is comprised in a different analyte (e.g., nucleic acid) from the first single-stranded target sequence.

#### Labelling Agents:

[0078] In some instances, provided herein are methods and compositions for analyzing endogenous analytes (e.g., RNA, ssDNA, and cell surface or intracellular proteins and/or metabolites) in a sample using one or more labelling agents. In some instances, an analyte labelling agent may include an agent that interacts with an analyte (e.g., an endogenous analyte in a sample). In some instances, the labelling agents can comprise a reporter oligonucleotide that is indicative of the analyte or portion thereof interacting with the labelling agent. For example, the reporter oligonucleotide may comprise a barcode sequence that permits identification of the labelling agent. In some cases, the sample contacted by the labelling agent can be further contacted with a probe (e.g., a single-stranded probe sequence), that hybridizes to a reporter oligonucleotide of the labelling agent, in order to identify the analyte associated with the labelling agent. In some

instances, the analyte labelling agent comprises an analyte binding moiety and a labelling agent barcode domain comprising one or more barcode sequences, e.g., a barcode sequence that corresponds to the analyte binding moiety and/or the analyte. An analyte binding moiety barcode includes to a barcode that is associated with or otherwise identifies the analyte binding moiety. In some instances, by identifying an analyte binding moiety by identifying its associated analyte binding moiety barcode, the analyte to which the analyte binding moiety binds can also be identified. An analyte binding moiety barcode can be a nucleic acid sequence of a given length and/or sequence that is associated with the analyte binding moiety. An analyte binding moiety barcode can generally include any of the variety of aspects of barcodes described herein.

[0079] In some instances, the method comprises one or more post-fixing (also referred to as post-fixation) steps after contacting the sample with one or more labelling agents.

[0080] In the methods and systems described herein, one or more labelling agents capable of binding to or otherwise coupling to one or more features may be used to characterize analytes, cells and/or cell features. In some instances, cell features include cell surface features. Analytes may include, but are not limited to, a protein, a receptor, an antigen, a surface protein, a transmembrane protein, a cluster of differentiation protein, a protein channel, a protein pump, a carrier protein, a phospholipid, a glycoprotein, a glycolipid, a cell-cell interaction protein complex, an antigen-presenting complex, a major histocompatibility complex, an engineered T-cell receptor, a T-cell receptor, a B-cell receptor, a chimeric antigen receptor, a gap junction, an adherens junction, or any combination thereof. In some instances, cell features may include intracellular analytes, such as proteins, protein modifications (e.g., phosphorylation status or other post-translational modifications), nuclear proteins, nuclear membrane proteins, or any combination thereof.

[0081] In some instances, an analyte binding moiety may include any molecule or moiety capable of binding to an analyte (e.g., a biological analyte, e.g., a macromolecular constituent). A labelling agent may include, but is not limited to, a protein, a peptide, an antibody (or an epitope binding fragment thereof), a lipophilic moiety (such as cholesterol), a cell surface receptor binding molecule, a receptor ligand, a small molecule, a bi-specific antibody, a bi-specific T-cell engager, a T-cell receptor engager, a B-cell receptor engager, a pro-body, an aptamer, a monobody, an affimer, a darpin, and a protein scaffold, or any combination thereof. The labelling agents can include (e.g., are attached to) a reporter oligonucleotide that is indicative of the cell surface feature to which the binding group binds. For example, the reporter oligonucleotide may comprise a barcode sequence that permits identification of the labelling agent. For example, a labelling agent that is specific to one type of cell feature (e.g., a first cell surface feature) may have coupled thereto a first reporter oligonucleotide, while a labelling agent that is specific to a different cell feature (e.g., a second cell surface feature) may have a different reporter oligonucleotide coupled thereto. For a description of exemplary labelling agents, reporter oligonucleotides, and methods of use, see, e.g., U.S. Pat. No. 10,550,429; U.S. Pat. Pub. 20190177800; and U.S. Pat. Pub. 20190367969, which are each incorporated by reference herein in their entirety.

[0082] In some instances, an analyte binding moiety includes one or more antibodies or antigen binding fragments thereof. The antibodies or antigen binding fragments including the analyte binding moiety can specifically bind to a target analyte. In some instances, the analyte is a protein (e.g., a protein on a surface of the biological sample (e.g., a cell) or an intracellular protein). In some instances, a plurality of analyte labelling agents comprising a plurality of analyte binding moieties bind a plurality of analytes present in a biological sample. In some instances, the plurality of analytes includes a single species of analyte (e.g., a single species of polypeptide). In some instances in which the plurality of analytes includes a single species of analyte, the analyte binding moieties of the plurality of analyte labelling agents are the same. In some instances in which the plurality of analytes includes a single species of analyte, the analyte binding moieties of the plurality of analyte labelling agents are the different (e.g., members of the plurality of analyte labelling agents can have two or more species of analyte binding moieties, wherein each of the two

or more species of analyte binding moieties binds a single species of analyte, e.g., at different binding sites). In some instances, the plurality of analytes includes multiple different species of analyte (e.g., multiple different species of polypeptides).

[0083] In other instances, e.g., to facilitate sample multiplexing, a labelling agent that is specific to a particular cell feature may have a first plurality of the labelling agent (e.g., an antibody or lipophilic moiety) coupled to a first reporter oligonucleotide and a second plurality of the labelling agent coupled to a second reporter oligonucleotide.

[0084] In some aspects, these reporter oligonucleotides may comprise nucleic acid barcode sequences that permit identification of the labelling agent which the reporter oligonucleotide is coupled to. The selection of oligonucleotides as the reporter may provide advantages of being able to generate significant diversity in terms of sequence, while also being readily attachable to most biomolecules, e.g., antibodies, etc., as well as being readily detected.

[0085] Attachment (coupling) of the reporter oligonucleotides to the labelling agents may be achieved through any of a variety of direct or indirect, covalent or non-covalent associations or attachments. For example, oligonucleotides may be covalently attached to a portion of a labelling agent (such a protein, e.g., an antibody or antibody fragment) using chemical conjugation techniques (e.g., Lightning-Link® antibody labelling kits available from Innova Biosciences), as well as other non-covalent attachment mechanisms, e.g., using biotinylated antibodies and oligonucleotides (or beads that include one or more biotinylated linker, coupled to oligonucleotides) with an avidin or streptavidin linker. Antibody and oligonucleotide biotinylation techniques are available. See, e.g., Fang, et al., “Fluoride-Cleavable Biotinylation Phosphoramidite for 5'-end-Labeling and Affinity Purification of Synthetic Oligonucleotides,” *Nucleic Acids Res.* Jan. 15, 2003; 31(2): 708-715, which is entirely incorporated herein by reference for all purposes. Likewise, protein and peptide biotinylation techniques have been developed and are readily available. See, e.g., U.S. Pat. No. 6,265,552, which is entirely incorporated herein by reference for all purposes. Furthermore, click reaction chemistry may be used to couple reporter oligonucleotides to labelling agents. Commercially available kits, such as those from Thunderlink and Abcam, and techniques common in the art may be used to couple reporter oligonucleotides to labelling agents as appropriate. In another example, a labelling agent is indirectly (e.g., via hybridization) coupled to a reporter oligonucleotide comprising a barcode sequence that identifies the label agent. For instance, the labelling agent may be directly coupled (e.g., covalently bound) to a hybridization oligonucleotide that comprises a sequence that hybridizes with a sequence of the reporter oligonucleotide. Hybridization of the hybridization oligonucleotide to the reporter oligonucleotide couples the labelling agent to the reporter oligonucleotide. In some instances, the reporter oligonucleotides are releasable from the labelling agent, such as upon application of a stimulus. For example, the reporter oligonucleotide may be attached to the labelling agent through a labile bond (e.g., chemically labile, photolabile, thermally labile, etc.) as generally described for releasing molecules from supports elsewhere herein. In some instances, the reporter oligonucleotides described herein may include one or more functional sequences that can be used in subsequent processing, such as an adapter sequence, a unique molecular identifier (UMI) sequence, a sequencer specific flow cell attachment sequence (such as an P5, P7, or partial P5 or P7 sequence), a primer or primer binding sequence, a sequencing primer or primer binding sequence (such as an R1, R2, or partial R1 or R2 sequence).

[0086] In some cases, the labelling agent can comprise a reporter oligonucleotide and a label. A label can be fluorophore, a radioisotope, a molecule capable of a colorimetric reaction, a magnetic particle, or any other suitable molecule or compound capable of detection. The label can be conjugated to a labelling agent (or reporter oligonucleotide) either directly or indirectly (e.g., the label can be conjugated to a molecule that can bind to the labelling agent or reporter oligonucleotide). In some cases, a label is conjugated to a first oligonucleotide that is complementary (e.g., hybridizes) to a sequence of the reporter oligonucleotide.

[0087] In some instances, multiple different species of analytes (e.g., polypeptides) from the biological sample can be subsequently associated with the one or more physical properties of the biological sample. For example, the multiple different species of analytes can be associated with locations of the analytes in the biological sample. Such information (e.g., proteomic information when the analyte binding moiety (ies) recognizes a polypeptide(s)) can be used in association with other spatial information (e.g., genetic information from the biological sample, such as DNA sequence information, transcriptome information (i.e., sequences of transcripts), or both). For example, a cell surface protein of a cell can be associated with one or more physical properties of the cell (e.g., a shape, size, activity, or a type of the cell). The one or more physical properties can be characterized by imaging the cell. The cell can be bound by an analyte labelling agent comprising an analyte binding moiety that binds to the cell surface protein and an analyte binding moiety barcode that identifies that analyte binding moiety. Results of protein analysis in a sample (e.g., a tissue sample or a cell) can be associated with DNA and/or RNA analysis in the sample.

#### Assays for In Situ Detection and Analysis:

[0088] Objectives for in situ detection and analysis methods include detecting, quantifying, and/or mapping analytes (e.g., gene activity) to specific regions in a biological sample (e.g., a tissue sample or cells deposited on a surface) at cellular or sub-cellular resolution. Methods for performing in situ studies include a variety of techniques, e.g., in situ hybridization and in situ sequencing techniques. These techniques allow one to study the subcellular distribution of target analytes (e.g., gene activity as evidenced, e.g., by expressed gene transcripts), and have the potential to provide crucial insights in the fields of developmental biology, oncology, immunology, histology, etc.

[0089] Various methods can be used for in situ detection and analysis of target analytes, e.g., sequencing by synthesis (SBS), sequencing by ligation (SBL), sequencing by hybridization (SBH). Non-limiting examples of in situ hybridization techniques include single molecule fluorescence in situ hybridization (smFISH) and multiplexed error-robust fluorescence in situ hybridization (MERFISH). smFISH enables in situ detection and quantification of gene transcripts in tissue samples at the locations where they reside by making use of libraries of multiple short oligonucleotide probes (e.g., approximately 20 base pairs (bp) in length), each labeled with a fluorophore. The probes are sequentially hybridized to gene sequences (e.g., DNA) or gene transcript sequences (e.g., mRNA) sequences, and visualized as diffraction-limited spots by fluorescence microscopy (Levsky, et al. (2003) "Fluorescence In situ Hybridization: Past, Present and Future", *Journal of Cell Science* 116(14): 2833-2838; Raj, et al. (2008) "Imaging Individual mRNA Molecules Using Multiple Singly Labeled Probes", *Nat Methods* 5(10): 877-879; Moor, et al. (2016), *ibid.*). Variations on the smFISH method include, for example, the use of combinatorial labelling schemes to improve multiplexing capability (Levsky, et al. (2003), *ibid.*), the use of smFISH in combination with super-resolution microscopy (Lubeck, et al. (2014) "Single-Cell In situ RNA Profiling by Sequential Hybridization", *Nature Methods* 11(4): 360-361).

[0090] MERFISH addresses two of the limitations of earlier in situ hybridization approaches, namely the limited number of target sequences that could be simultaneously identified and the robustness of the approach to readout errors caused by the stochastic nature of the hybridization process (Moor, et al. (2016), *ibid.*). MERFISH utilizes a binary barcoding scheme in which the probed target mRNA sequences are either fluorescence positive or fluorescence negative for any given imaging cycle (Ke, et al. (2016), *ibid.*; Moffitt, et al. (2016) "RNA Imaging with Multiplexed Error Robust Fluorescence In situ Hybridization", *Methods Enzymol.* 572:1-49). The encoding probes that contain a combination of target-specific hybridization sequence regions and barcoded readout sequence regions are first hybridized to the target mRNA sequences. In each imaging cycle, a subset of fluorophore-conjugated readout probes is hybridized to a subset of encoding probes. Target mRNA sequences that fluoresce in a given cycle are assigned a value of "1" and the remaining target mRNA sequences are assigned a value of "0". Between imaging cycles, the



fluorescent probes from the previous cycle are photobleached. After, e.g., 14 or 16 rounds of readout probe hybridization and imaging, unique combinations of the detected fluorescence signals generate a 14-bit or 16-bit code that identifies the different gene transcripts. To address the increased error rate for correctly calling the readout codes increases as the number of hybridization and imaging cycles increases, the method may also entail the use of Hamming distances for barcode design and correction of decoding errors (see., e.g., Buschmann, et al. (2013) “Levenshtein Error-Correcting Barcodes for Multiplexed DNA Sequencing”, *Bioinformatics* 14:272), thereby resulting in an error-robust barcoding scheme.

[0091] Some in situ sequencing techniques generally comprise both in situ target capture (e.g., of mRNA sequences) and in situ sequencing. Non-limiting examples of in situ sequencing techniques include in situ sequencing with padlock probes (ISS-PLP), fluorescent in situ sequencing (FISSEQ), barcode in situ targeted sequencing (Barista-Seq), and spatially-resolved transcript amplicon readout mapping (STARmap) (see, e.g., Ke, et al. (2016), *ibid.*, Asp, et al. (2020), *ibid.*).

[0092] Some methods for in situ detection and analysis of analytes utilize a probe (e.g., padlock or circular probe) that detects specific target analytes. The in situ sequencing using padlock probes (ISS-PLP) method, for example, combines padlock probing to target specific gene transcripts, rolling-circle amplification (RCA), and sequencing by ligation (SBL) chemistry. Within intact tissue sections, reverse transcription primers are hybridized to target sequence (e.g., mRNA sequences) and reverse transcription is performed to create cDNA to which a padlock probe (a single-stranded DNA molecule comprising regions that are complementary to the target cDNA) can bind (see, e.g., Asp, et al. (2020), *ibid.*). In one variation of the method, the padlock probe binds to the cDNA target with a gap remaining between the ends which is then filled in using a DNA polymerization reaction. In another variation of the method, the ends of the bound padlock probe are adjacent to each other. The ends are then ligated to create a circular DNA molecule. Target amplification using rolling-circle amplification (RCA) results in micrometer-sized RCA products (RCPs), containing a plurality of concatenated repeats of the probe sequence. In some examples, RCPs are then subjected to, e.g., sequencing-by-ligation (SBL) or sequencing-by-hybridization (SBH). In some cases, the method allows for a barcode located within the probe to be decoded.

Products of Endogenous Analytes and/or Labelling Agents:

[0093] In some instances, provided herein are methods and compositions for analyzing one or more products of an endogenous analyte and/or a labelling agent in a biological sample. In some instances, an endogenous analyte (e.g., a viral or cellular DNA or RNA) or a product (e.g., a hybridization product, a ligation product, an extension product (e.g., by a DNA or RNA polymerase), a replication product, a transcription/reverse transcription product, and/or an amplification product such as a rolling circle amplification (RCA) product) thereof is analyzed. In some instances, a labelling agent that directly or indirectly binds to an analyte in the biological sample is analyzed. In some instances, a product (e.g., a hybridization product, a ligation product, an extension product (e.g., by a DNA or RNA polymerase), a replication product, a transcription/reverse transcription product, and/or an amplification product such as a rolling circle amplification (RCA) product) of a labelling agent that directly or indirectly binds to an analyte in the biological sample is analyzed.

[0094] In some instances, the analyzing comprises using primary probes which comprise a target binding region (e.g., a region that binds to a target such as RNA transcripts) and the primary probes may contain one or more barcodes (e.g., primary barcode). In some instances, the barcodes are bound by detection primary probes, which do not need to be fluorescent, but that include a target-binding portion (e.g., for hybridizing to one or more primary probes) and one or more barcodes (e.g., secondary barcodes). In some instances, the detection primary probe comprises an overhang that does not hybridize to the target nucleic acid but hybridizes to another probe. In some examples, the overhang comprises the barcode(s). In some instances, the barcodes of the detection primary probes are targeted by detectably labeled detection oligonucleotides, such as fluorescently labeled

oligos. In some instances, one or more decoding schemes are used to decode the signals, such as fluorescence, for sequence determination. Various probes and probe sets can be used to hybridize to and detect an endogenous analyte and/or a sequence associated with a labelling agent. In some instances, these assays may enable multiplexed detection, signal amplification, combinatorial decoding, and error correction schemes. Exemplary barcoded probes or probe sets may be based on a padlock probe, a gapped padlock probe, a SNAIL (Splint Nucleotide Assisted Intramolecular Ligation) probe set, a PLAYR (Proximity Ligation Assay for RNA) probe set, a PLISH (Proximity Ligation in situ Hybridization) probe set. The specific probe or probe set design can vary.

#### Hybridization and Ligation:

[0095] Various probes and probe sets can be hybridized to an endogenous analyte and/or a labelling agent and each probe may comprise one or more barcode sequences. The specific probe or probe set design can vary. In some instances, the hybridization of a primary probe or probe set (e.g., a circularizable probe or probe set) to a target nucleic acid analyte and may lead to the generation of a rolling circle amplification (RCA) template. In some instances, the assay uses or generates a circular nucleic acid molecule which can be the RCA template.

[0096] In some instances, a product of an endogenous analyte and/or a labelling agent is a ligation product. In some instances, the ligation product is formed from circularization of a circularizable probe or probe set upon hybridization to a target sequence. In some instances, the ligation product is formed between two or more endogenous analytes. In some instances, the ligation product is formed between an endogenous analyte and a labelling agent. In some instances, the ligation product is formed between two or more labelling agent. In some instances, the ligation product is an intramolecular ligation of an endogenous analyte. In some instances, the ligation product is an intramolecular ligation of a labelling agent, for example, the circularization of a circularizable probe or probe set upon hybridization to a target sequence. The target sequence can be comprised in an endogenous analyte (e.g., nucleic acid such as a genomic DNA or mRNA) or a product thereof (e.g., cDNA from a cellular mRNA transcript), or in a labelling agent (e.g., the reporter oligonucleotide) or a product thereof.

[0097] In some instances, provided herein is a probe or probe set capable of DNA-templated ligation, such as from a cDNA molecule. Sec, e.g., U.S. Pat. No. 8,551,710, which is hereby incorporated by reference in its entirety. In some instances, provided herein is a probe or probe set capable of RNA-templated ligation. Sec, e.g., U.S. Pat. Pub. 2020/0224244 which is hereby incorporated by reference in its entirety. In some instances, the probe set is a SNAIL probe set. See, e.g., U.S. Pat. Pub. 20190055594, which is hereby incorporated by reference in its entirety. In some instances, provided herein is a multiplexed proximity ligation assay. Sec, e.g., U.S. Pat. Pub. 20140194311 which is hereby incorporated by reference in its entirety. In some instances, provided herein is a probe or probe set capable of proximity ligation, for instance a proximity ligation assay for RNA (e.g., PLAYR) probe set. See, e.g., U.S. Pat. Pub. 20160108458, which is hereby incorporated by reference in its entirety. In some instances, a circular probe can be indirectly hybridized to the target nucleic acid. In some instances, the circular construct is formed from a probe set capable of proximity ligation, for instance a proximity ligation in situ hybridization (PLISH) probe set. Sec, e.g., U.S. Pat. Pub. 2020/0224243 which is hereby incorporated by reference in its entirety.

[0098] In some instances, the ligation involves chemical ligation. In some instances, the ligation involves template dependent ligation. In some instances, the ligation involves template independent ligation. In some instances, the ligation involves enzymatic ligation.

[0099] In some instances, the enzymatic ligation involves use of a ligase. In some aspects, the ligase used herein comprises an enzyme that is commonly used to join polynucleotides together or to join the ends of a single polynucleotide. An RNA ligase, a DNA ligase, or another variety of ligase can be used to ligate two nucleotide sequences together. Ligases comprise ATP-dependent double-strand polynucleotide ligases, NAD-i-dependent double-strand DNA or RNA ligases and

single-strand polynucleotide ligases, for example any of the ligases described in EC 6.5.1.1 (ATP-dependent ligases), EC 6.5.1.2 (NAD<sup>+</sup>-dependent ligases), EC 6.5.1.3 (RNA ligases). Specific examples of ligases comprise bacterial ligases such as *E. coli* DNA ligase, Tth DNA ligase, *Thermococcus* sp. (strain 9° N) DNA ligase (9° N<sup>TM</sup> DNA ligase, New England Biolabs), Taq DNA ligase, Ampligase<sup>TM</sup> (Epicentre Biotechnologies) and phage ligases such as T3 DNA ligase, T4 DNA ligase and T7 DNA ligase and mutants thereof. In some instances, the ligase is a T4 RNA ligase. In some instances, the ligase is a splintR ligase. In some instances, the ligase is a single stranded DNA ligase. In some instances, the ligase is a T4 DNA ligase. In some instances, the ligase is a ligase that has an DNA-splinted DNA ligase activity. In some instances, the ligase is a ligase that has an RNA-splinted DNA ligase activity.

[0100] In some instances, the ligation herein is a direct ligation. In some instances, the ligation herein is an indirect ligation. “Direct ligation” means that the ends of the polynucleotides hybridize immediately adjacently to one another to form a substrate for a ligase enzyme resulting in their ligation to each other (intramolecular ligation). Alternatively, “indirect” means that the ends of the polynucleotides hybridize non-adjacently to one another, i.e., separated by one or more intervening nucleotides or “gap”. In some instances, said ends are not ligated directly to each other, but instead occurs either via the intermediacy of one or more intervening (so-called “gap” or “gap-filling” (oligo) nucleotides) or by the extension of 3′ end of a probe to “fill” the “gap” corresponding to said intervening nucleotides (intermolecular ligation). In some cases, the gap of one or more nucleotides between the hybridized ends of the polynucleotides may be “filled” by one or more “gap” (oligo) nucleotide(s) which are complementary to a splint, padlock probe, or target nucleic acid. The gap may be a gap of 1 to 60 nucleotides or a gap of 1 to 40 nucleotides or a gap of 3 to 40 nucleotides. In specific implementations, the gap may be a gap of about 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 or more nucleotides, of any integer (or range of integers) of nucleotides in between the indicated values. In some instances, the gap between said terminal regions may be filled by a gap oligonucleotide or by extending 3′ end of a polynucleotide. In some cases, ligation involves ligating the ends of the probe to at least one gap (oligo) nucleotide, such that the gap (oligo) nucleotide becomes incorporated into the resulting polynucleotide. In some instances, the ligation herein is preceded by gap filling. In other implementations, the ligation herein does not require gap filling.

[0101] In some instances, ligation of the polynucleotides produces polynucleotides with melting temperature higher than that of un-ligated polynucleotides. Thus, in some aspects, ligation stabilizes the hybridization complex containing the ligated polynucleotides prior to subsequent steps, comprising amplification and detection.

[0102] In some aspects, a high fidelity ligase, such as a thermostable DNA ligase (e.g., a Taq DNA ligase), is used. Thermostable DNA ligases are active at elevated temperatures, allowing further discrimination by incubating the ligation at a temperature near the melting temperature of the DNA strands. This selectively reduces the concentration of annealed mismatched substrates (expected to have a slightly lower T<sub>m</sub> around the mismatch) over annealed fully base-paired substrates. Thus, high-fidelity ligation can be achieved through a combination of the intrinsic selectivity of the ligase active site and balanced conditions to reduce the incidence of annealed mismatched dsDNA.

[0103] In some instances, the ligation herein is a proximity ligation of ligating two (or more) nucleic acid sequences that are in proximity with each other, e.g., through enzymatic means (e.g., a ligase). In some instances, proximity ligation can include a “gap-filling” step that involves incorporation of one or more nucleic acids by a polymerase, based on the nucleic acid sequence of a template nucleic acid molecule, spanning a distance between the two nucleic acid molecules of interest (see, e.g., U.S. Pat. No. 7,264,929, the entire contents of which are incorporated herein by reference). A wide variety of different methods can be used for proximity ligating nucleic acid molecules, including (but not limited to) “sticky-end” and “blunt-end” ligations. Additionally, single-stranded ligation can be used to perform proximity ligation on a single-stranded nucleic acid molecule. Sticky-end proximity ligations involve the hybridization of complementary single-

stranded sequences between the two nucleic acid molecules to be joined, prior to the ligation event itself. Blunt-end proximity ligations generally do not include hybridization of complementary regions from each nucleic acid molecule because both nucleic acid molecules lack a single-stranded overhang at the site of ligation.

#### Primer Extension and Amplification:

[0104] In some instances, the hybridization of a primary probe or probe set (e.g. a circularizable probe or probe set) to a target analyte and may lead to the generation of an extension or amplification product. In some instances, a product is a primer extension product of an analyte, a labelling agent, a probe or probe set bound to the analyte (e.g., a circularizable probe bound to genomic DNA, mRNA, or cDNA), or a probe or probe set bound to the labelling agent (e.g., a circularizable probe bound to one or more reporter oligonucleotides from the same or different labelling agents).

[0105] A primer is generally a single-stranded nucleic acid sequence having a 3' end that can be used as a substrate for a nucleic acid polymerase in a nucleic acid extension reaction. RNA primers are formed of RNA nucleotides, and are used in RNA synthesis, while DNA primers are formed of DNA nucleotides and used in DNA synthesis. Primers can also include both RNA nucleotides and DNA nucleotides (e.g., in a random or designed pattern). Primers can also include other natural or synthetic nucleotides described herein that can have additional functionality. In some examples, DNA primers can be used to prime RNA synthesis and vice versa (e.g., RNA primers can be used to prime DNA synthesis). Primers can vary in length. For example, primers can be about 6 bases to about 120 bases. For example, primers can include up to about 25 bases. A primer, may in some cases, refer to a primer binding sequence. A primer extension reaction generally refers to any method where two nucleic acid sequences become linked (e.g., hybridized) by an overlap of their respective terminal complementary nucleic acid sequences (i.e., for example, 3' termini). Such linking can be followed by nucleic acid extension (e.g., an enzymatic extension) of one, or both termini using the other nucleic acid sequence as a template for extension. Enzymatic extension can be performed by an enzyme including, but not limited to, a polymerase and/or a reverse transcriptase.

[0106] In some instances, a product of an endogenous analyte and/or a labelling agent is an amplification product of one or more polynucleotides, for instance, a circular probe or circularizable probe or probe set. In some instances, the disclosed methods may comprise the use of a rolling circle amplification (RCA) technique to amplify signal. Rolling circle amplification is an isothermal, DNA polymerase-mediated process in which long single-stranded DNA molecules are synthesized on a short circular single-stranded DNA template using a single DNA primer (Zhao, et al. (2008), "Rolling Circle Amplification: Applications in Nanotechnology and Biodetection with Functional Nucleic Acids", *Angew Chem Int Ed Engl.* 47(34): 6330-6337; Ali, et al. (2014), "Rolling Circle Amplification: A Versatile Tool for Chemical Biology, Materials Science and Medicine", *Chem Soc Rev.* 43(10): 3324-3341). The RCA product is a concatemer containing tens to hundreds of tandem repeats that are complementary to the circular template, and may be used to develop sensitive techniques for the detection of a variety of targets, including nucleic acids (DNA, RNA), small molecules, proteins, and cells (Ali, et al. (2014), *ibid.*). In some implementations, a primer that hybridizes to the circular probe or circularized probe is added and used as such for amplification. In some instances, the RCA comprises a linear RCA, a branched RCA, a dendritic RCA, or any combination thereof.

[0107] In some instances, the amplification is performed at a temperature between or between about 20° C. and about 60° C. In some instances, the amplification is performed at a temperature between or between about 30° C. and about 40° C. In some aspects, the amplification step, such as the rolling circle amplification (RCA) is performed at a temperature between at or about 25° C. and at or about 50° C., such as at or about 25° C., 27° C., 29° C., 31° C., 33° C., 35° C., 37° C., 39° C., 41° C., 43° C., 45° C., 47° C., or 49° C.

[0108] In some instances, upon addition of a DNA polymerase in the presence of appropriate dNTP precursors and other cofactors, a primer is elongated to produce multiple copies of the circular template. This amplification step can utilize isothermal amplification or non-isothermal amplification. In some instances, after the formation of the hybridization complex and association of the amplification probe, the hybridization complex is rolling-circle amplified to generate a cDNA nanoball (i.e., amplicon) containing multiple copies of the cDNA. Techniques for rolling circle amplification (RCA) are known in the art such as linear RCA, a branched RCA, a dendritic RCA, or any combination thereof. (See, e.g., Baner et al, Nucleic Acids Research, 26:5073-5078, 1998; Lizardi et al, Nature Genetics 19:226, 1998; Mohsen et al., Acc Chem Res. 2016 Nov. 15; 49(11): 2540-2550; Schweitzer et al. Proc. Natl Acad. Sci. USA 97:101 13-1 19, 2000; Faruqi et al, BMC Genomics 2:4, 2000; Nallur et al, Nucl. Acids Res. 29:el 18, 2001; Dean et al. Genome Res. 11:1095-1099, 2001; Schweitzer et al, Nature Biotech. 20:359-365, 2002; U.S. Pat. Nos. 6,054,274, 6,291,187, 6,323,009, 6,344,329 and 6,368,801). Exemplary polymerases for use in RCA comprise DNA polymerase such as phi29 ( $\phi$ 29) polymerase, Klenow fragment, *Bacillus stearothermophilus* DNA polymerase (BST), T4 DNA polymerase, T7 DNA polymerase, or DNA polymerase I. In some aspects, DNA polymerases that have been engineered or mutated to have desirable characteristics can be employed. In some instances, the polymerase is phi29 DNA polymerase.

[0109] In some aspects, during the amplification step, modified nucleotides can be added to the reaction to incorporate the modified nucleotides in the amplification product (e.g., nanoball). Exemplary of the modified nucleotides comprise amine-modified nucleotides. In some aspects of the methods, for example, for anchoring or cross-linking of the generated amplification product (e.g., nanoball) to a scaffold, to cellular structures and/or to other amplification products (e.g., other nanoballs). In some aspects, the amplification products comprises a modified nucleotide, such as an amine-modified nucleotide. In some instances, the amine-modified nucleotide comprises an acrylic acid N-hydroxysuccinimide moiety modification. Examples of other amine-modified nucleotides comprise, but are not limited to, a 5-Aminoallyl-dUTP moiety modification, a 5-Propargylamino-dCTP moiety modification, a N6-6-Aminohexyl-dATP moiety modification, or a 7-Deaza-7-Propargylamino-dATP moiety modification.

[0110] In some instances, the RCA template may comprise the target analyte, or a part thereof, where the target analyte is a nucleic acid, or it may be provided or generated as a proxy, or a marker, for the analyte. In some instances, the RCA template may comprise a sequence of the probes and probe sets hybridized to an endogenous analyte and/or a labelling agent. In some instances, the amplification product can be generated as a proxy, or a marker, for the analyte. As noted above, many assays are known for the detection of numerous different analytes, which use a RCA-based detection system, e.g., where the signal is provided by generating a RCP from a circular RCA template which is provided or generated in the assay, and the RCP is detected to detect the analyte. The RCP may thus be regarded as a reporter which is detected to detect the target analyte. However, the RCA template may also be regarded as a reporter for the target analyte; the RCP is generated based on the RCA template, and comprises complementary copies of the RCA template. The RCA template determines the signal which is detected, and is thus indicative of the target analyte. As will be described in more detail below, the RCA template may be a probe, or a part or component of a probe, or may be generated from a probe, or it may be a component of a detection assay (i.e. a reagent in a detection assay), which is used as a reporter for the assay, or a part of a reporter, or signal-generation system. The RCA template used to generate the RCP may thus be a circular (e.g. circularized) reporter nucleic acid molecule, namely from any RCA-based detection assay which uses or generates a circular nucleic acid molecule as a reporter for the assay. Since the RCA template generates the RCP reporter, it may be viewed as part of the reporter system for the assay.

[0111] In some instances, an assay may detect a product herein that includes a molecule or a

complex generated in a series of reactions, e.g., hybridization, ligation, extension, replication, transcription/reverse transcription, and/or amplification (e.g., rolling circle amplification), in any suitable combination. For example, a product comprising a target sequence for a probe disclosed herein (e.g., a bridge probe or L-probe) may be a hybridization complex formed of a cellular nucleic acid in a sample and an exogenously added nucleic acid probe. The exogenously added nucleic acid probe may comprise an overhang that does not hybridize to the cellular nucleic acid but hybridizes to another probe (e.g., a detection probe). The exogenously added nucleic acid probe may be optionally ligated to a cellular nucleic acid molecule or another exogenous nucleic acid molecule. In other examples, a product comprising a target sequence for a probe disclosed herein (e.g., an anchor probe) may be an RCP of a circularizable probe or probe set which hybridizes to a cellular nucleic acid molecule (e.g., genomic DNA or mRNA) or product thereof (e.g., a transcript such as cDNA, a DNA-templated ligation product of two probes, or an RNA-templated ligation product of two probes). In other examples, a product comprising a target sequence for a probe disclosed herein (e.g., a bridge probe or L-probe) may be a probe hybridizing to an RCP. The probe may comprise an overhang that does not hybridize to the RCP but hybridizes to another probe (e.g., a detection probe).

#### Signal Amplification Methods:

[0112] In some instances, a method disclosed herein may also comprise one or more signal amplification components and detecting such signals. In some instances, the present disclosure relates to the detection of nucleic acid sequences in situ using probe hybridization and generation of amplified signals associated with the probes. In some instances, the target nucleic acid of a nucleic acid probe comprises multiple target sequences for nucleic acid probe hybridization, such that the signal corresponding to a barcode sequence of the nucleic acid probe is amplified by the presence of multiple nucleic acid probes hybridized to the target nucleic acid. For example, multiple sequences can be selected from a target nucleic acid such as an mRNA, such that a group of nucleic acid probes (e.g., 20-50 nucleic acid probes) hybridize to the mRNA in a tiled fashion. In another example, the target nucleic acid can be an amplification product (e.g., an RCA product) comprising multiple copies of a target sequence (e.g., a barcode sequence of the RCA product).

[0113] Alternatively or additionally, amplification of a signal associated with a barcode sequence of a nucleic acid probe can be amplified using one or more signal amplification strategies off of an oligonucleotide probe that hybridizes to the barcode sequence. In some aspects, amplification of the signal associated with the oligonucleotide probe can reduce the number of nucleic acid probes needed to hybridize to the target nucleic acid to obtain a sufficient signal-to-noise ratio. For example, the number of nucleic acid probes to tile a target nucleic acid such as an mRNA can be reduced. In some aspects, reducing the number of nucleic acid probes tiling a target nucleic acid enables detection of shorter target nucleic acids, such as shorter mRNAs. In some instances, no more than one, two, three, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleic acid probes may be hybridized to the target nucleic acid. In instances wherein the target nucleic acid is an amplification product, signal amplification off of the oligonucleotide probes may reduce the number of target sequences required for detection (e.g., the length of the RCA product can be reduced).

[0114] Exemplary signal amplification methods include targeted deposition of detectable reactive molecules around the site of probe hybridization, targeted assembly of branched structures (e.g., bDNA or branched assay using locked nucleic acid (LNA)), programmed in situ growth of concatemers by enzymatic rolling circle amplification (RCA) (e.g., as described in US 2019/0055594 incorporated herein by reference), hybridization chain reaction, assembly of topologically catenated DNA structures using serial rounds of chemical ligation (clampFISH), signal amplification via hairpin-mediated concatemerization (e.g., as described in US 2020/0362398 incorporated herein by reference), e.g., primer exchange reactions such as signal amplification by exchange reaction (SABER) or SABER with DNA-Exchange (Exchange-

SABER). In some instances, a non-enzymatic signal amplification method may be used.

[0115] The detectable reactive molecules may comprise tyramide, such as used in tyramide signal amplification (TSA) or multiplexed catalyzed reporter deposition (CARD)-FISH. In some instances, the detectable reactive molecule may be releasable and/or cleavable from a detectable label such as a fluorophore. In some instances, a method disclosed herein comprises multiplexed analysis of a biological sample comprising consecutive cycles of probe hybridization, fluorescence imaging, and signal removal, where the signal removal comprises removing the fluorophore from a fluorophore-labeled reactive molecule (e.g., tyramide). Exemplary detectable reactive reagents and methods are described in U.S. Pat. No. 6,828,109, US 2019/0376956, WO 2019/236841, WO 2020/102094, WO 2020/163397, and WO 2021/067475, all of which are incorporated herein by reference in their entireties.

[0116] In some instances, hybridization chain reaction (HCR) can be used for signal amplification. HCR is an enzyme-free nucleic acid amplification based on a triggered chain of hybridization of nucleic acid molecules starting from HCR monomers, which hybridize to one another to form a nicked nucleic acid polymer. This polymer is the product of the HCR reaction which is ultimately detected in order to indicate the presence of the target analyte. HCR is described in detail in Dirks and Pierce, 2004, PNAS, 101(43), 15275-15278 and in U.S. Pat. Nos. 7,632,641 and 7,721,721 (see also US 2006/00234261; Chemeris et al, 2008 Doklady Biochemistry and Biophysics, 419, 53-55; Niu et al, 2010, 46, 3089-3091; Choi et al, 2010, Nat. Biotechnol. 28(11), 1208-1212; and Song et al, 2012, Analyst, 137, 1396-1401). HCR monomers typically comprise a hairpin, or other metastable nucleic acid structure. In the simplest form of HCR, two different types of stable hairpin monomer, referred to here as first and second HCR monomers, undergo a chain reaction of hybridization events to form a long nicked double-stranded DNA molecule when an “initiator” nucleic acid molecule is introduced. The HCR monomers have a hairpin structure comprising a double stranded stem region, a loop region connecting the two strands of the stem region, and a single stranded region at one end of the double stranded stem region. The single stranded region which is exposed (and which is thus available for hybridization to another molecule, e.g. initiator or other HCR monomer) when the monomers are in the hairpin structure may be known as the “toehold region” (or “input domain”). The first HCR monomers each further comprise a sequence which is complementary to a sequence in the exposed toehold region of the second HCR monomers. This sequence of complementarity in the first HCR monomers may be known as the “interacting region” (or “output domain”). Similarly, the second HCR monomers each comprise an interacting region (output domain), e.g. a sequence which is complementary to the exposed toehold region (input domain) of the first HCR monomers. In the absence of the HCR initiator, these interacting regions are protected by the secondary structure (e.g. they are not exposed), and thus the hairpin monomers are stable or kinetically trapped (also referred to as “metastable”), and remain as monomers (e.g. preventing the system from rapidly equilibrating), because the first and second sets of HCR monomers cannot hybridize to each other. However, once the initiator is introduced, it is able to hybridize to the exposed toehold region of a first HCR monomer, and invade it, causing it to open up. This exposes the interacting region of the first HCR monomer (e.g. the sequence of complementarity to the toehold region of the second HCR monomers), allowing it to hybridize to and invade a second HCR monomer at the toehold region. This hybridization and invasion in turn opens up the second HCR monomer, exposing its interacting region (which is complementary to the toehold region of the first HCR monomers), and allowing it to hybridize to and invade another first HCR monomer. The reaction continues in this manner until all of the HCR monomers are exhausted (e.g. all of the HCR monomers are incorporated into a polymeric chain). Ultimately, this chain reaction leads to the formation of a nicked chain of alternating units of the first and second monomer species. The presence of the HCR initiator is thus required in order to trigger the HCR reaction by hybridization to and invasion of a first HCR monomer. The first and second HCR monomers are designed to hybridize to one another and are thus may be defined as cognate to one

another. They are also cognate to a given HCR initiator sequence. HCR monomers which interact with one another (hybridize) may be described as a set of HCR monomers or an HCR monomer, or hairpin, system.

[0117] An HCR reaction could be carried out with more than two species or types of HCR monomers. For example, a system involving three HCR monomers could be used. In such a system, each first HCR monomer may comprise an interacting region which binds to the toehold region of a second HCR monomer; each second HCR may comprise an interacting region which binds to the toehold region of a third HCR monomer; and each third HCR monomer may comprise an interacting region which binds to the toehold region of a first HCR monomer. The HCR polymerization reaction would then proceed as described above, except that the resulting product would be a polymer having a repeating unit of first, second and third monomers consecutively. Corresponding systems with larger numbers of sets of HCR monomers could readily be conceived. Branching HCR systems have also been devised and described (see, e.g., WO 2020/123742 incorporated herein by reference), and may be used in the methods herein.

[0118] In some instances, similar to HCR reactions that use hairpin monomers, linear oligo hybridization chain reaction (LO-HCR) can also be used for signal amplification. In some instances, provided herein is a method of detecting an analyte in a sample comprising: (i) performing a linear oligo hybridization chain reaction (LO-HCR), wherein an initiator is contacted with a plurality of LO-HCR monomers of at least a first and a second species to generate a polymeric LO-HCR product hybridized to a target nucleic acid molecule, wherein the first species comprises a first hybridization region complementary to the initiator and a second hybridization region complementary to the second species, wherein the first species and the second species are linear, single-stranded nucleic acid molecules; wherein the initiator is provided in one or more parts, and hybridizes directly or indirectly to or is comprised in the target nucleic acid molecule; and (ii) detecting the polymeric product, thereby detecting the analyte. In some instances, the first species and/or the second species may not comprise a hairpin structure. In some instances, the plurality of LO-HCR monomers may not comprise a metastable secondary structure. In some instances, the LO-HCR polymer may not comprise a branched structure. In some instances, performing the linear oligo hybridization chain reaction comprises contacting the target nucleic acid molecule with the initiator to provide the initiator hybridized to the target nucleic acid molecule. In any of the instances herein, the target nucleic acid molecule and/or the analyte can be an RCA product.

[0119] In some instances, detection of nucleic acids sequences in situ includes combination of the sequential decoding methods described herein with an assembly for branched signal amplification. In some instances, the assembly complex comprises an amplifier hybridized directly or indirectly (via one or more oligonucleotides) to a sequence of an oligonucleotide probe described herein. In some instances, the assembly includes one or more amplifiers each including an amplifier repeating sequence. In some aspects, the one or more amplifiers is labeled. Described herein is a method of using the aforementioned assembly, including for example, using the assembly in multiplexed error-robust fluorescent in situ hybridization (MERFISH) applications, with branched DNA amplification for signal readout. In some instances, the amplifier repeating sequence is about 5-30 nucleotides, and is repeated N times in the amplifier. In some instances, the amplifier repeating sequence is about 20 nucleotides, and is repeated at least two times in the amplifier. In some aspects, the one or more amplifier repeating sequence is labeled. For exemplary branched signal amplification, see e.g., U.S. Pat. Pub. No. US20200399689A1 and Xia et al., Multiplexed Detection of RNA using MERFISH and branched DNA amplification. Scientific Reports (2019), each of which is fully incorporated by reference herein.

[0120] In some instances, an oligonucleotide probe described herein can be associated with an amplified signal by a method that comprises signal amplification by performing a primer exchange reaction (PER). In various instances, a primer with domain on its 3' end binds to a catalytic hairpin,



and is extended with a new domain by a strand displacing polymerase. For example, a primer with domain 1 on its 3' ends binds to a catalytic hairpin, and is extended with a new domain 1 by a strand displacing polymerase, with repeated cycles generating a concatemer of repeated domain 1 sequences. In various instances, the strand displacing polymerase is Bst. In various instances, the catalytic hairpin includes a stopper which releases the strand displacing polymerase. In various instances, branch migration displaces the extended primer, which can then dissociate. In various instances, the primer undergoes repeated cycles to form a concatemer primer (see e.g., U.S. Pat. Pub. No. US20190106733, which is incorporated herein by reference, for exemplary molecules and PER reaction components).

#### Barcoded Analytes and Detection:

[0121] A target sequence for a probe disclosed herein may be comprised in any analyte disclosed herein, including an endogenous analyte (e.g., a viral or cellular nucleic acid), a labelling agent, or a product generated in the biological sample using an endogenous analyte and/or a labelling agent.

[0122] In some aspects, one or more of the target sequences includes or is associated with one or more barcode(s), e.g., at least two, three, four, five, six, seven, eight, nine, ten, or more barcodes. Barcodes can spatially-resolve molecular components found in biological samples, for example, within a cell or a tissue sample. A barcode can be attached to an analyte or to another moiety or structure in a reversible or irreversible manner. A barcode can be added to, for example, a fragment of a deoxyribonucleic acid (DNA) or ribonucleic acid (RNA) sample before or during sequencing of the sample. Barcodes can allow for identification and/or quantification of individual sequencing-reads (e.g., a barcode can be or can include a unique molecular identifier or "UMI"). In some aspects, a barcode comprises about 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or more than 30 nucleotides.

[0123] In some instances, a barcode includes two or more sub-barcodes that together function as a single barcode. For example, a polynucleotide barcode can include two or more polynucleotide sequences (e.g., sub-barcodes) that are separated by one or more non-barcode sequences. In some instances, the one or more barcode(s) can also provide a platform for targeting functionalities, such as oligonucleotides, oligonucleotide-antibody conjugates, oligonucleotide-streptavidin conjugates, modified oligonucleotides, affinity purification, detectable moieties, enzymes, enzymes for detection assays or other functionalities, and/or for detection and identification of the polynucleotide.

[0124] In any of the preceding implementations, barcodes (e.g., primary and/or secondary barcode sequences) can be analyzed (e.g., detected or sequenced) using any suitable method or technique, including those described herein, such as sequencing by synthesis (SBS), sequencing by ligation (SBL), or sequencing by hybridization (SBH). In some instances, barcoding schemes and/or barcode detection schemes as described in RNA sequential probing of targets (RNA SPOTs), single-molecule fluorescent in situ hybridization (smFISH), multiplexed error-robust fluorescence in situ hybridization (MERFISH) or sequential fluorescence in situ hybridization (seqFISH+) can be used. In any of the preceding implementations, the methods provided herein can include analyzing the barcodes by sequential hybridization and detection with a plurality of labelled probes (e.g., detection probes (e.g., detection oligos) or barcode probes). In some instances, the barcode detection steps can be performed as described in hybridization-based in situ sequencing (HybISS). In some instances, probes can be detected and analyzed (e.g., detected or sequenced) as performed in fluorescent in situ sequencing (FISSEQ), or as performed in the detection steps of the spatially-resolved transcript amplicon readout mapping (STARmap) method. In some instances, signals associated with an analyte can be detected as performed in sequential fluorescent in situ hybridization (seqFISH).

[0125] In some instances, in a barcode-based detection method, barcode sequences are detected for identification of other molecules including nucleic acid molecules (DNA or RNA) longer than the barcode sequences themselves, as opposed to direct sequencing of the longer nucleic acid

molecules. In some instances, a N-mer barcode sequence comprises  $4^N$  complexity given a sequencing read of N bases, and a much shorter sequencing read may be required for molecular identification compared to non-barcode sequencing methods such as direct sequencing. For example, 1024 molecular species may be identified using a 5-nucleotide barcode sequence ( $4^{5.5}=1024$ ), whereas 8 nucleotide barcodes can be used to identify up to 65,536 molecular species, a number greater than the total number of distinct genes in the human genome. In some instances, the barcode sequences contained in the probes or RCPs are detected, rather than endogenous sequences, which can be an efficient read-out in terms of information per cycle of sequencing. Because the barcode sequences are pre-determined, they can also be designed to feature error detection and correction mechanisms, see, e.g., U.S. Pat. Pub. 20190055594 and WO2019199579A1, which are hereby incorporated by reference in their entirety.

#### Sequential Hybridization:

[0126] In some instances, the present disclosure relates to methods and compositions for encoding and detecting analytes in a temporally sequential manner for in situ analysis of an analyte in a biological sample, e.g., a target nucleic acid in a cell in an intact tissue. In some aspects, provided herein is a method for detecting the detectably-labeled probes, thereby generating a signal signature. In some instances, the signal signature corresponds to an analyte of the plurality of analytes. In some instances, the methods described herein are based, in part, on the development of a multiplexed biological assay and readout, in which a sample is first contacted with a plurality of nucleic acid probes comprising one or more probe types (e.g., labelling agent, circularizable probe, circular probe, etc.), allowing the probes to directly or indirectly bind target analytes, which may then be optically detected (e.g., by detectably-labeled probes) in a temporally-sequential manner. In some instances, the probes or probe sets comprising various probe types may be applied to a sample simultaneously. In some instances, the probes or probe sets comprising various probe types may be applied to a sample sequentially. In some aspects, the method comprises sequential hybridization of labelled probes to create a spatiotemporal signal signature or code that identifies the analyte.

[0127] In some aspects, provided herein is a method involving a multiplexed biological assay and readout, in which a sample is first contacted with a plurality of nucleic acid probes, allowing the probes to directly or indirectly bind target analytes, which may then be optically detected (e.g., by detectably-labeled probes) in a temporally sequential manner. The plurality of nucleic acid probes themselves may be detectably-labeled and detected; in other words, the nucleic acid probes themselves serve as the detection probes. In other implementations, a nucleic acid probe itself is not directly detectably-labeled (e.g., the probe itself is not conjugated to a detectable label); rather, in addition to a target binding sequence (e.g., a sequence binding to a barcode sequence in an RCA product), the nucleic acid probe further comprises a sequence for detection which can be recognized by one or more detectably-labeled detection probes. In some instances, the probes or probe sets comprising various probe types may be applied to a sample simultaneously. In some instances, the probes or probe sets comprising various probe types may be applied to a sample sequentially. In some instances, the method comprises detecting a plurality of analytes in a sample.

[0128] In some instances, the method presented herein comprises contacting the sample with a plurality of probes comprising one or more probes having distinct labels and detecting signals from the plurality of probes in a temporally sequential manner, wherein said detection generates signal signatures each comprising a temporal order of signal or absence thereof, and the signal signatures correspond to said plurality of probes that identify the corresponding analytes. In some instances, the temporal order of the signals or absence thereof corresponding to the analytes can be unique for each different analyte of interest in the sample. In some instances, the plurality of probes hybridize to an endogenous molecule in the sample, such as a cellular nucleic acid molecule, e.g., genomic DNA, RNA (e.g., mRNA), or cDNA. In some instances, the plurality of probes hybridize to a product of an endogenous molecule in the sample (e.g., directly or indirectly via an intermediate

probe). In some instances, the plurality of probes hybridize to labelling agent that binds directly or indirectly to an endogenous molecule in the sample or a product thereof. In some instances, the plurality of probes hybridize to a product (e.g., an RCA product) of a labelling agent that binds directly or indirectly to an endogenous molecule in the sample or a product thereof.

[0129] In any of the implementations disclosed herein, the detection of signals can be performed sequentially in cycles, one for each distinct label. In any of the implementations disclosed herein, signals or absence thereof from detectably-labeled probes targeting an analyte in a particular location in the sample can be recorded in a first cycle for detecting a first label, and signals or absence thereof from detectably-labeled probes targeting the analyte in the particular location can be recorded in a second cycle for detecting a second label distinct from the first label. In any of the implementations disclosed herein, a unique signal signature can be generated for each analyte of the plurality of analytes. In any of the implementations disclosed herein, one or more molecules comprising the same analyte or a portion thereof can be associated with the same signal signature.

[0130] In some instances, the in situ assays employ strategies for optically encoding the spatial location of target analytes (e.g., mRNAs) in a sample using sequential rounds of fluorescent hybridization. Microscopy may be used to analyze 4 or 5 fluorescent colors indicative of the spatial localization of a target, followed by various rounds of hybridization and stripping, in order to generate a large set of unique optical signal signatures assigned to different analytes. These methods often require a large number of hybridization rounds, and a large number of microscope lasers (e.g., detection channels) to detect a large number of fluorophores, resulting in a one to one mapping of the lasers to the fluorophores. Specifically, each detectably-labeled probe comprises one detectable moiety, e.g., a fluorophore.

[0131] In some aspects, provided herein is a method for analyzing a sample using a detectably-labeled set of probes. In some instances, the method comprises contacting the sample with a first plurality of detectably-labeled probes for targeting a plurality of analytes; performing a first detection round comprising detecting signals from the first plurality of detectably-labeled probes; contacting the sample with a second plurality of detectably-labeled probes for targeting the plurality of analytes; performing a second detection round of detecting signals from the second plurality of detectably-labeled probes, thereby generating a signal signature comprising a plurality of signals detected from the first detection round and second detection round, wherein the signal signature corresponds to an analyte of the plurality of analytes.

[0132] In some instances, detection of an optical signal signature comprises several rounds of detectably-labeled probe hybridization (e.g., contacting a sample with detectably-labeled probes), detectably-labeled probe detection, and detectably-labeled probe removal. In some instances, a sample is contacted with plurality first detectably-labeled probes, and said probes are hybridized to a plurality of nucleic acid analytes within the sample in decoding hybridization round 1. In some instances, a first detection round is performed following detectably-labeled probe hybridization. After hybridization and detection of a first plurality of detectably-labeled probes, probes are removed, and a sample may be contacted with a second plurality round of detectably-labeled probes targeting the analytes targeted in decoding hybridization round 1. The second plurality of detectably-labeled probes may hybridize to the same nucleic acid(s) as the first plurality of detectably-labeled probes (e.g., hybridize to an identical or hybridize to new nucleic acid sequence within the same nucleic acid), or the second plurality of detectably-labeled probes may hybridize to different nucleic acid(s) compared to the first plurality of detectably-labeled probes. Following m rounds of contacting a sample with a plurality of detectably-labeled probes, probe detection, and probe removal, ultimately a unique signal signature to each nucleic acid is produced that may be used to identify and quantify said nucleic acids and the corresponding analytes (e.g., if the nucleic acids themselves are not the analytes of interest and each is used as part of a labelling agent for one or more other analytes such as protein analytes and/or other nucleic acid analytes).

[0133] In some instances, after hybridization of a detectably-labeled probes (e.g., fluorescently

labeled oligonucleotide) that detects a sequence (e.g., barcode sequence on a secondary probe or a primary probe), and optionally washing away the unbound molecules of the detectably-labeled probe, the sample is imaged and the detection oligonucleotide or detectable label is inactivated and/or removed. In some instances, removal of the signal associated with the hybridization between rounds can be performed by washing, heating, stripping, enzymatic digestion, photo-bleaching, displacement (e.g., displacement of detectably-labeled probes with another reagent or nucleic acid sequence), cleavage, quenching, chemical degradation, bleaching, oxidation, or any combinations thereof.

[0134] In some examples, removal of a probe (e.g., un-hybridizing the entire probe), signal modifications (e.g., quenching, masking, photo-bleaching, signal enhancement (e.g., via FRET), signal amplification, etc.), signal removal (e.g., cleaving off or permanently inactivating a detectable label) can be performed. Inactivation may be caused by removal of the detectable label (e.g., from the sample, or from the probe, etc.), and/or by chemically altering the detectable label in some fashion, e.g., by photobleaching the detectable label, bleaching or chemically altering the structure of the detectable label, e.g., by reduction, etc.). In some instances, the fluorescently labeled oligonucleotide and/or the intermediate probe hybridized to the fluorescently labeled oligonucleotide (e.g., bridge probe or L-probe) can be removed. In some instances, a fluorescent detectable label may be inactivated by chemical or optical techniques such as oxidation, photobleaching, chemically bleaching, stringent washing or enzymatic digestion or reaction by exposure to an enzyme, dissociating the detectable label from other components (e.g., a probe), chemical reaction of the detectable label (e.g., to a reactant able to alter the structure of the detectable label) or the like. For instance, bleaching may occur by exposure to oxygen, reducing agents, or the detectable label could be chemically cleaved from the nucleic acid probe and washed away via fluid flow.

[0135] In some instances, removal of a signal comprises displacement of probes with another reagent (e.g., probe) or nucleic acid sequence. For example, a given probe (e.g., detectably-labeled probes and/or the intermediate probe hybridized to the fluorescently labeled oligonucleotide (e.g., bridge probe or L-probe)) may be displaced by a subsequent probe that hybridizes to an overlapping region shared between the binding sites of the probes. In some cases, a displacement reaction can be very efficient, and thus allows for probes to be switched quickly between cycles, without the need for chemical stripping (or any of the damage to the sample that is associated therewith). In some instances, a sequence for hybridizing the subsequent or displacer probe (i.e. a toehold sequence) may be common across a plurality of probes capable of hybridizing to a given binding site. In some aspects, a single displacement probe can be used to simultaneously displace detection probes bound to an equivalent barcode position from all of the RCPs within a given sample simultaneously (with the displacement mediated by the subsequent detection probes). This may further increase efficiency and reduce the cost of the method, as fewer different probes are required.

[0136] After a signal is inactivated and/or removed, then the sample is re-hybridized in a subsequent round with a subsequent fluorescently labeled oligonucleotide, and the oligonucleotide can be labeled with the same color or a different color as the fluorescently labeled oligonucleotide of the previous cycle. In some instances, as the positions of the analytes, probes, and/or products thereof can be fixed (e.g., via fixing and/or crosslinking) in a sample, the fluorescent spot corresponding to an analyte, probe, or product thereof remains in place during multiple rounds of hybridization and can be aligned to read out a string of signals associated with each target analyte. Decoding:

[0137] A “decoding process” is a process comprising a plurality of decoding cycles in which different sets of barcode probes are contacted with target analytes (e.g., mRNA sequences) or target barcodes (e.g., barcodes associated with target analytes) present in a sample, and used to detect the target sequences or associated target barcodes, or segments thereof. In some instances, the

decoding process comprises acquiring one or more images (e.g., fluorescence images) for each decoding cycle. Decoded barcode sequences are then inferred based on a set of physical signals (e.g., fluorescence signals) detected in each decoding cycle of a decoding process. In some instances, the set of physical signals (e.g., fluorescence signals) detected in a series of decoding cycles for a given target barcode (or target analyte sequence) may be considered a “signal signature” for the target barcode (or target analyte sequence). In some instances, a decoding process may comprise, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more than 10 decoding cycles. In some instances, each decoding cycle may comprise contacting a plurality of target sequences or target barcodes with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more than 10 barcode probes (e.g., fluorescently-labeled barcode probes) that are configured to hybridize or bind to specific target sequences or target barcodes, or segments thereof. In some instances, a decoding process may comprise performing a series of in situ barcode probe hybridization steps and acquiring images (e.g., fluorescence images) at each step. Systems and methods for performing multiplexed fluorescence in situ hybridization and imaging are described in, for example, WO 2021/127019 A1; U.S. Pat. No. 11,021,737; and PCT/EP2020/065090 (WO2020240025A1), each of which is incorporated herein by reference in its entirety.

#### Anchor Probes:

[0138] In some instances, the present methods may further involve contacting the target analyte, e.g., a nucleic acid molecule, or proxy thereof with an anchor probe. In some instances, the anchor probe comprises a sequence complementary to an anchor probe binding region, which is present in all target nucleic acid molecules (e.g., in primary or secondary probes), and a detectable label. The detection of the anchor probe via the detectable label confirms the presence of the target nucleic acid molecule. The target nucleic acid molecule may be contacted with the anchor probe prior to, concurrently with, or after being contacted with the first set of detection probes. In some instances, the target nucleic acid molecule may be contacted with the anchor probe during multiple decoding cycles. In some instances, multiple different anchor probes comprising different sequences and/or different reporters may be used to confirm the presence of multiple different target nucleic acid molecules. The use of multiple anchor probes is particularly useful when detection of a large number of target nucleic acid molecules is required, as it allows for optical crowding to be reduced and thus for detected target nucleic acid molecules to be more clearly resolved

[0139] Described herein is a method applying the aforementioned techniques to reduce optical crowding in an assay. In some instances, a method can include analyzing a biological sample, by contacting the biological sample with a first number of primary probe(s) configured to hybridize to a first target nucleic acid and a second number of primary probes configured to hybridize to a second target nucleic acid, each primary probe including a target-hybridizing region configured to hybridize to a different target region in the corresponding target nucleic acid, and a barcode region, and the first number is 1 or more, and the second number is greater than the first number, contacting the biological sample with a plurality of detectable probes, wherein each detectable probe is configured to hybridize to a barcode sequence in the barcode regions of the first number of primary probe(s) and/or the second number of primary probes, or a complement of the barcode sequence, detecting a signal associated with the plurality of detectable probes or absence thereof at one or more locations in the biological sample, and contacting the biological sample with a subsequent plurality of detectable probes, wherein each detectable probe in the subsequent plurality is configured to hybridize to a subsequent barcode sequence in the barcode regions of the first number of primary probe(s) and/or the second number of primary probes, or a complement of the subsequent barcode sequence. In some instances, wherein the second number is greater than the first number, the difference is based on absolute or relative numbers of the first target nucleic acid and second target nucleic acid in a sample. In some instances, the difference between second and first number is linearly or non-linearly proportional to absolute or relative numbers of the first target nucleic acid and second target nucleic acid in a sample. In some instances, the difference is

inversely proportional, where the first number is for first target nucleic acid present in greater amounts (i.e., abundant), and the greater second number is for second target nucleic acid present in lesser amounts.

[0140] Reference will now be made in detail to implementations and embodiments of various aspects and variations of systems and methods described herein. Although several exemplary variations of the systems and methods are described herein, other variations of the systems and methods may include aspects of the systems and methods described herein combined in any suitable manner having combinations of all or some of the aspects described.

[0141] Target molecules (e.g., nucleic acids, proteins, antibodies, etc.) can be detected in biological samples (e.g., one or more cells or a tissue sample) using an instrument having integrated optics and fluidics modules (an “opto-fluidic instrument” or “opto-fluidic system”). In an opto-fluidic instrument, the fluidics module is configured to deliver one or more reagents (e.g., fluorescent probes) to the biological sample and/or remove spent reagents therefrom. Additionally, the optics module is configured to illuminate the biological sample with light having one or more spectral emission curves (over a range of wavelengths) and subsequently capture one or more images of emitted light signals from the biological sample during one or more probing cycles. In various embodiments, the captured images may be processed in real time and/or at a later time to determine the presence of the one or more target molecules in the biological sample, as well as three-dimensional position information associated with each detected target molecule. Additionally, the opto-fluidics instrument includes a sample module configured to receive (and, optionally, secure) one or more biological samples. In some instances, the sample module includes an X-Y stage configured to move the biological sample along an X-Y plane (e.g., perpendicular to an objective lens of the optics module).

[0142] In various embodiments, the opto-fluidic instrument is configured to analyze one or more target molecules in their naturally occurring place (i.e., in situ) within the biological sample. For example, an opto-fluidic instrument may be an in-situ analysis system used to analyze a biological sample and detect target molecules including but not limited to DNA, RNA, proteins, antibodies, and/or the like.

[0143] A sample disclosed herein can be or be derived from any biological sample. Biological samples may be obtained from any suitable source using any of a variety of techniques including, but not limited to, biopsy, surgery, and laser capture microscopy (LCM), and generally includes cells, tissues, and/or other biological material from the subject. A biological sample can be obtained from a prokaryote such as a bacterium, an archaea, a virus, or a viroid. A biological sample can also be obtained from non-mammalian organisms (e.g., a plant, an insect, an arachnid, a nematode, a fungus, or an amphibian). A biological sample can also be obtained from a eukaryote, such as a tissue sample from a mammal. A biological sample from an organism may comprise one or more other organisms or components therefrom. For example, a mammalian tissue section may comprise a prion, a viroid, a virus, a bacterium, a fungus, or components from other organisms, in addition to mammalian cells and non-cellular tissue components. Subjects from which biological samples can be obtained can be healthy or asymptomatic subjects, subjects that have or are suspected of having a disease (e.g., an individual with a disease such as cancer) or a pre-disposition to a disease, and/or subjects in need of therapy or suspected of needing therapy.

[0144] The biological sample can include any number of macromolecules, for example, cellular macromolecules and organelles (e.g., mitochondria and nuclei). The biological sample can be obtained as a tissue sample, such as a tissue section, biopsy, a core biopsy, needle aspirate, or fine needle aspirate. The sample can be a fluid sample, such as a blood sample, urine sample, or saliva sample. The sample can be a skin sample, a colon sample, a cheek swab, a histology sample, a histopathology sample, a plasma or serum sample, a tumor sample, living cells, cultured cells, a clinical sample such as, for example, whole blood or blood-derived products, blood cells, or cultured tissues or cells, including cell suspensions.

[0145] In some embodiments, the biological sample may comprise cells or a tissue sample which are deposited on a substrate. As described herein, a substrate can be any support that is insoluble in aqueous liquid and allows for positioning of biological samples, analytes, features, and/or reagents on the support. In some embodiments, a biological sample is attached to a substrate. In some embodiments, the substrate is optically transparent to facilitate analysis on the opto-fluidic instruments disclosed herein. For example, in some instances, the substrate is a glass substrate (e.g., a microscopy slide, cover slip, or other glass substrate). Attachment of the biological sample can be irreversible or reversible, depending upon the nature of the sample and subsequent steps in the analytical method. In certain embodiments, the sample can be attached to the substrate reversibly by applying a suitable polymer coating to the substrate and contacting the sample to the polymer coating. The sample can then be detached from the substrate, e.g., using an organic solvent that at least partially dissolves the polymer coating. Hydrogels are examples of polymers that are suitable for this purpose. In some embodiments, the substrate can be coated or functionalized with one or more substances to facilitate attachment of the sample to the substrate. Suitable substances that can be used to coat or functionalize the substrate include, but are not limited to, lectins, polylysine, antibodies, and polysaccharides.

[0146] It is to be noted that, although the above discussion relates to an opto-fluidic instrument that can be used for in situ target molecule detection via probe hybridization, the discussion herein equally applies to any opto-fluidic instrument that employs any imaging or target molecule detection technique. That is, for example, an opto-fluidic instrument may include a fluidics module that includes fluids needed for establishing the experimental conditions required for the probing of target molecules in the sample. Further, such an opto-fluidic instrument may also include a sample module configured to receive the sample, and an optics module including an imaging system for illuminating (e.g., exciting one or more fluorescent probes within the sample) and/or imaging light signals received from the probed sample. The in-situ analysis system may also include other ancillary modules configured to facilitate the operation of the opto-fluidic instrument, such as, but not limited to, cooling systems, motion calibration systems, etc.

[0147] FIG. 3 shows an example workflow of analysis of a biological sample **110** (e.g., cell or tissue sample) using an opto-fluidic instrument **120**, according to various embodiments. In various embodiments, the sample **110** can be a biological sample (e.g., a tissue) that includes molecules such as DNA, RNA, proteins, antibodies, etc. For example, the sample **110** can be a sectioned tissue that is treated to access the RNA thereof for labeling with circularizable DNA probes. Ligation of the probes may generate a circular DNA probe which can be enzymatically amplified and bound with fluorescent oligonucleotides, which can create bright signal that is convenient to image and has a high signal-to-noise ratio.

[0148] In various embodiments, the sample **110** may be placed in the opto-fluidic instrument **120** for analysis and detection of the molecules in the sample **110**. In various embodiments, the opto-fluidic instrument **120** can be a system configured to facilitate the experimental conditions conducive for the detection of the target molecules. For example, the opto-fluidic instrument **120** can include a fluidics module **140**, an optics module **150**, a sample module **160**, and an ancillary module **170**, and these modules may be operated by a system controller **130** to create the experimental conditions for the probing of the molecules in the sample **110** by selected probes (e.g., circularizable DNA probes), as well as to facilitate the imaging of the probed sample (e.g., by an imaging system of the optics module **150**). In various embodiments, the various modules of the opto-fluidic instrument **120** may be separate components in communication with each other, or at least some of them may be integrated together.

[0149] In various embodiments, the sample module **160** may be configured to receive the sample **110** into the opto-fluidic instrument **120**. For instance, the sample module **160** may include a sample interface module (SIM) that is configured to receive a sample device (e.g., cassette) onto which the sample **110** can be deposited. That is, the sample **110** may be placed in the opto-fluidic

instrument **120** by depositing the sample **110** (e.g., the sectioned tissue) on a sample device that is then inserted into the SIM of the sample module **160**. In some instances, the sample module **160** may also include an X-Y stage onto which the SIM is mounted. The X-Y stage may be configured to move the SIM mounted thereon (e.g., and as such the sample device containing the sample **110** inserted therein) in perpendicular directions along the two-dimensional (2D) plane of the opto-fluidic instrument **120**.

[0150] The experimental conditions that are conducive for the detection of the molecules in the sample **110** may depend on the target molecule detection technique that is employed by the opto-fluidic instrument **120**. For example, in various embodiments, the opto-fluidic instrument **120** can be a system that is configured to detect molecules in the sample **110** via hybridization of probes. In such cases, the experimental conditions can include molecule hybridization conditions that result in the intensity of hybridization of the target molecule (e.g., nucleic acid) to a probe (e.g., oligonucleotide) being significantly higher when the probe sequence is complementary to the target molecule than when there is a single-base mismatch. The hybridization conditions include the preparation of the sample **110** using reagents such as washing/stripping reagents, hybridizing reagents, etc., and such reagents may be provided by the fluidics module **140**.

[0151] In various embodiments, the fluidics module **140** may include one or more components that may be used for storing the reagents, as well as for transporting said reagents to and from the sample device containing the sample **110**. For example, the fluidics module **140** may include reservoirs configured to store the reagents, as well as a waste container configured for collecting the reagents (e.g., and other waste) after use by the opto-fluidic instrument **120** to analyze and detect the molecules of the sample **110**. Further, the fluidics module **140** may also include pumps, tubes, pipettes, etc., that are configured to facilitate the transport of the reagent to the sample device (e.g., and as such the sample **110**). For instance, the fluidics module **140** may include pumps (“reagent pumps”) that are configured to pump washing/stripping reagents to the sample device for use in washing/stripping the sample **110** (e.g., as well as other washing functions such as washing an objective lens of the imaging system of the optics module **150**).

[0152] In various embodiments, the ancillary module **170** can be a cooling system of the opto-fluidic instrument **120**, and the cooling system may include a network of coolant-carrying tubes that are configured to transport coolants to various modules of the opto-fluidic instrument **120** for regulating the temperatures thereof. In such cases, the fluidics module **140** may include coolant reservoirs for storing the coolants and pumps (e.g., “coolant pumps”) for generating a pressure differential, thereby forcing the coolants to flow from the reservoirs to the various modules of the opto-fluidic instrument **120** via the coolant-carrying tubes. In some instances, the fluidics module **140** may include returning coolant reservoirs that may be configured to receive and store returning coolants, i.e., heated coolants flowing back into the returning coolant reservoirs after absorbing heat discharged by the various modules of the opto-fluidic instrument **120**. In such cases, the fluidics module **140** may also include cooling fans that are configured to force air (e.g., cool and/or ambient air) into the returning coolant reservoirs to cool the heated coolants stored therein. In some instance, the fluidics module **140** may also include cooling fans that are configured to force air directly into a component of the opto-fluidic instrument **120** so as to cool said component. For example, the fluidics module **140** may include cooling fans that are configured to direct cool or ambient air into the system controller **130** to cool the same.

[0153] As discussed above, the opto-fluidic instrument **120** may include an optics module **150** which include the various optical components of the opto-fluidic instrument **120**, such as but not limited to a camera, an illumination module (e.g., light source such as LEDs), an objective lens, and/or the like. The optics module **150** may include a fluorescence imaging system that is configured to image the fluorescence emitted by the probes (e.g., oligonucleotides) in the sample **110** after the probes are excited by light from the illumination module of the optics module **150**.

[0154] In some instances, the optics module **150** may also include an optical frame onto which the



camera, the illumination module, and/or the X-Y stage of the sample module **160** may be mounted. [0155] In various embodiments, the system controller **130** may be configured to control the operations of the opto-fluidic instrument **120** (e.g., and the operations of one or more modules thereof). In some instances, the system controller **130** may take various forms, including a processor, a single computer (or computer system), or multiple computers in communication with each other. In various embodiments, the system controller **130** may be communicatively coupled with data storage, set of input devices, display system, or a combination thereof. In some cases, some or all of these components may be considered to be part of or otherwise integrated with the system controller **130**, may be separate components in communication with each other, or may be integrated together. In other examples, the system controller **130** can be, or may be in communication with, a cloud computing platform.

[0156] In various embodiments, the opto-fluidic instrument **120** may analyze the sample **110** and may generate the output **190** that includes indications of the presence of the target molecules in the sample **110**. For instance, with respect to the example embodiment discussed above where the opto-fluidic instrument **120** employs a hybridization technique for detecting molecules, the opto-fluidic instrument **120** may cause the sample **110** to undergo successive rounds of fluorescent probe hybridization (using two or more sets of fluorescent probes, where each set of fluorescent probes is excited by a different color channel) and be imaged to detect target molecules in the probed sample **110**. In such cases, the output **190** may include optical signatures (e.g., a codeword) specific to each gene, which allow the identification of the target molecules.

[0157] In some instances, an assembly for transilluminating a substrate can include a sample carrier device (e.g., a microfluidic chip or glass slide), a thermal control module configured to control the temperature of the sample carrier device (e.g., a thermoelectric module), and a light source configured to illuminate the sample carrier device. In some instances, the assembly includes a heat exchanger (e.g., a fluid block having a cooling fluid flowing therethrough). In some instances, an assembly for transilluminating can include sample carrier device (e.g., a sample substrate), an optically transparent substrate, a light source configured to illuminate the optically transparent substrate, a light scattering layer configured to scatter light from the light source, and/or a thermal control module configured to control the temperature of the sample carrier device and/or optically transparent substrate.

[0158] In some embodiments, the sample carrier device (e.g., a cassette) can be configured to receive a sample. In some embodiments, the sample carrier device can include one or more microfluidic channels, e.g., sample chambers or microfluidic channels etched into a planar substrate or chambers within a flow cell or microfluidic device.

[0159] A sample carrier device for the systems disclosed herein can include, but is not limited to, a substrate configured to receive a sample, a microscope slide and/or an adapter configured to mount microscope slides (with or without coverslips) on a microscope stage or automated stage (e.g., an automated translation or rotational stage), a substrate, and/or an adapter configured to mount slides on a microscope stage or automated stage, a substrate comprising etched sample containment chambers (e.g., chambers open to the environment) and/or an adapter configured to mount such substrates on a microscope stage or automated stage, a flow cell and/or an adapter configured to mount flow cells on a microscope stage or automated stage, or a microfluidic device and/or an adapter configured to mount microfluidic devices on a microscope stage or automated stage. In some embodiments, the sample carrier device further includes a cassette configured to secure a substrate (e.g., a glass slide). In some embodiments, the cassette includes two or more components (e.g., a top half and a bottom half) into which the substrate is secured.

[0160] In some instances, the one or more sample carrier devices can be designed for performing a variety of chemical analysis, biochemical analysis, nucleic acid analysis, cell analysis, or tissue analysis applications. In some instances, for example, the sample carrier device (e.g., flow cells and microfluidic devices) may comprise a sample, e.g., a tissue sample. In some instances, the sample

carrier device (e.g., flow cells and microfluidic devices) may comprise a sample, e.g., a tissue sample, placed in contact with, e.g., a substrate (e.g., a surface of the flow cell or microfluidic device).

[0161] The sample carrier devices for the disclosed systems (e.g., microscope slides, substrates comprising one or more etched microfluidic channel, flow cells or microfluidic devices comprising one or more microfluidic channels, etc.) can be fabricated from any of a variety of materials known to those of skill in the art including, but not limited to, glass (e.g., borosilicate glass, soda lime glass, etc.), fused silica (quartz), silicon, polymer (e.g., polystyrene (PS), macroporous polystyrene (MPPS), polymethylmethacrylate (PMMA), polycarbonate (PC), polypropylene (PP), polyethylene (PE), high density polyethylene (HDPE), cyclic olefin polymers (COP), cyclic olefin copolymers (COC), polyethylene terephthalate (PET), polydimethylsiloxane (PDMS), etc.), polyetherimide (PEI) and perfluoroelastomer (FFKM) as more chemically inert alternatives, or any combination thereof. FFKM is also known as Kalrez.

[0162] The one or more materials used to fabricate sample carrier devices for the disclosed systems (e.g., substrates configured to receive a sample, microscope slides, substrates comprising one or more etched microfluidic channels, flow cells or microfluidic devices comprising one or more microfluidic channels or sample chambers, etc.) can be optically transparent to facilitate use with spectroscopic or imaging-based detection techniques. In some instances, the entire sample carrier device can be optically transparent. Alternatively, in some instances, only a portion of the sample carrier device (e.g., an optically transparent “window”) can be optically transparent.

[0163] The sample carrier devices for the disclosed systems (e.g., substrates configured to receive a sample, microscope slides, substrates comprising one or more etched microfluidic channels, flow cells or microfluidic devices comprising one or more microfluidic channels or sample chambers, etc.) can be fabricated using any of a variety of techniques known to those of skill in the art, where the choice of fabrication technique is often dependent on the choice of material used, and vice versa. Examples of suitable sample carrier device fabrication techniques include, but are not limited to, extrusion, drawing, precision computer numerical control (CNC) machining and boring, laser photoablation, photolithography in combination with wet chemical etching, deep reactive ion etching (DRIE), micro-molding, embossing, 3D-printing, thermal bonding, adhesive bonding, anodic bonding, and the like (see, e.g., Gale, et al. (2018), “A Review of Current Methods in Microfluidic Device Fabrication and Future Commercialization Prospects”, *Inventions* 3, 60, 1-25, which is hereby incorporated by reference in its entirety).

[0164] FIG. 4A illustrates a cross-sectional view of an optics module **200** in an imaging system. One or more illumination sources **210**, e.g., one or more light emitting diodes (LEDs), provides light through one or more optical components and an objective lens **220** to thereby illuminate a sample **250**. In various embodiments, the optical components include a collimator **211**. In various embodiments, the optical components include a field stop **212**. In various embodiments, the optical components include one or more excitation filters **213**. In various embodiments, the one or more excitation filters **213** are configured to filter light from the illumination source(s) **210** for a predetermined range of wavelengths (e.g., each filter has one or more blocking band(s) and/or transmission band(s) that may be different or may overlap at least in part) and each excitation filter **213** is aligned with appropriate illumination sources (e.g., blue LEDs, green LEDs, yellow LEDs, red LEDs, ultraviolet LEDs, etc.). In various embodiments, the optical components include a condenser **214**. In various embodiments, the optical components include a beam splitter **215**. An optical axis **251** is illustrated extending through the center of the optical surfaces in the objective lens **220** and its path includes an image plane, a focal plane, and input/output pupils (illustrated in FIG. 4B).

[0165] A sensor array **260** (e.g., CMOS sensor) receives light signals from the sample **250**. In various embodiments, the optical components include one or more emission filters **265**. In various embodiments, the one or more emission filters **265** are configured to filter light from the sample

(e.g., emitted from one or more fluorophores, autofluorescence, etc.) for a predetermined range of wavelengths (e.g., each filter has one or more blocking band(s) and/or transmission band(s) that may be different or may overlap at least in part). In various embodiments, the emission filters **265** align (e.g., via motorized translation) with optics and/or the sensor array. In various embodiments, the sample **250** is probed with fluorescent probes configured to bind to a target (e.g., DNA or RNA) that, when illuminated with a particular wavelength (or range of wavelengths) of light, emit light signals that can be detected by the sensor array **260**. In various embodiments, the sample **250** is repeatedly probed with two or more (e.g., two, three, four, five, six, etc.) different sets of probes. In various embodiments, each set of probes corresponds to a specific color (e.g., blue, green, yellow, or red) such that, when illuminated by that color, probes bound to a target emit light signals. In some embodiments, the sensor array **260** is aligned with the optical axis **251** of the objective lens **220** (i.e., the optical axis of the camera is coincident with and parallel to the optical axis of the objective lens **220**). In various embodiments, the sensor array **260** is positioned perpendicularly to the objective lens **220** (i.e., the optical axis of the camera is perpendicular to and intersects the optical axis of the objective lens **220**). In various embodiments, a tube lens **261** is mounted in the optical path to focus light on the sensor array **260** thereby allowing for image formation with infinity-corrected objectives. Descriptions of optical modules and illumination assemblies for use in opto-fluidic instruments can be found in U.S. provisional patent application No. 63/427,282, filed on Nov. 22, 2022, titled “Systems and Methods for Illuminating a Sample” and U.S. provisional patent application No. 63/427,360, file on Nov. 22, 2022, titled “Systems and Methods for Imaging Samples,” each of which is incorporated by reference in its entirety.

[0166] In various embodiments, the sample is illuminated with one or more wavelengths configured to induce fluorescence in the sample. In various embodiments, the sample is probed during one or more probing cycles with one or more fluorescent probes configured to bind to one or more target analytes. In various embodiments, the one or more wavelengths are selected to induce fluorescence in a subset of the one or more fluorescent probes. In various embodiments, each probing cycle includes illumination with two or more (e.g., four) colors of light. In various embodiments, the sample is treated with a fluorescent stain configured to illuminate one or more structures within the sample. In various embodiments, the sample is contacted with a nuclear stain. In various embodiments, the sample is contacted with 4',6-diamidino-2-phenylindole (“DAPI”) configured to bind to adenine-thymine-rich regions in DNA. In various embodiments, illumination of the sample causes autofluorescence of the sample. In various embodiments, autofluorescence is the natural emission of light by biological structures when they have absorbed light, and may be used to distinguish the light originating from artificially added fluorescent markers. In various embodiments, fluorescence of the sample through fluorescent probes, autofluorescence, and/or a fluorescent stain can be used with the methods described herein to determine one or more focus metrics of a tissue sample.

[0167] In various embodiments, the sample is illuminated via edge lighting or transillumination along one or more edges of the sample and/or sample substrate. In various embodiments, the edge lighting provides dark-field illumination of the sample. In various embodiments, edge lighting is provided by one or more light sources positioned to provide light substantially perpendicular to a normal of the substrate surface on which the sample is disposed. In various embodiments, the substrate is a glass slide. In various embodiments, the substrate is configured as a wave guide to thereby guide light emitted from the edge lighting towards the sample. In various embodiments, illumination of the sample via edge lighting can be used with the methods described herein to determine one or more focus metrics of a tissue sample.

[0168] Referring now to FIG. 5, a schematic of an example of a computing node is shown. Computing node **10** is only one example of a suitable computing node and is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the invention described herein. Regardless, computing node **10** is capable of being implemented and/or

performing any of the functionality set forth hereinabove.

[0169] In computing node **10** there is a computer system/server **12**, which is operational with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with computer system/server **12** include, but are not limited to, personal computer systems, server computer systems, thin clients, thick clients, handheld or laptop devices, multiprocessor systems, microprocessor-based systems, set top boxes, programmable consumer electronics, network PCs, minicomputer systems, mainframe computer systems, and distributed cloud computing environments that include any of the above systems or devices, and the like.

[0170] Computer system/server **12** may be described in the general context of computer system-executable instructions, such as program modules, being executed by a computer system.

Generally, program modules may include routines, programs, objects, components, logic, data structures, and so on that perform particular tasks or implement particular abstract data types. Computer system/server **12** may be practiced in distributed cloud computing environments where tasks are performed by remote processing devices that are linked through a communications network. In a distributed cloud computing environment, program modules may be located in both local and remote computer system storage media including memory storage devices.

[0171] As shown in FIG. 5, computer system/server **12** in computing node **10** is shown in the form of a general-purpose computing device. The components of computer system/server **12** may include, but are not limited to, one or more processors or processing units **16**, a system memory **28**, and a bus **18** that couples various system components including system memory **28** to processor **16**.

[0172] Bus **18** represents one or more of any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus.

[0173] Computer system/server **12** typically includes a variety of computer system readable media. Such media may be any available media that is accessible by computer system/server **12**, and it includes both volatile and non-volatile media, removable and non-removable media.

[0174] System memory **28** can include computer system readable media in the form of volatile memory, such as random access memory (RAM) **30** and/or cache memory **32**. Computer system/server **12** may further include other removable/non-removable, volatile/non-volatile computer system storage media. By way of example only, storage system **34** can be provided for reading from and writing to a non-removable, non-volatile magnetic media (not shown and typically called a “hard drive”). Although not shown, a magnetic disk drive for reading from and writing to a removable, non-volatile magnetic disk (e.g., a “floppy disk”), and an optical disk drive for reading from or writing to a removable, non-volatile optical disk such as a CD-ROM, DVD-ROM or other optical media can be provided. In such instances, each can be connected to bus **18** by one or more data media interfaces. As will be further depicted and described below, memory **28** may include at least one program product having a set (e.g., at least one) of program modules that are configured to carry out the functions of embodiments of the invention.

[0175] Program/utility **40**, having a set (at least one) of program modules **42**, may be stored in memory **28** by way of example, and not limitation, as well as an operating system, one or more application programs, other program modules, and program data. Each of the operating system, one or more application programs, other program modules, and program data or some combination thereof, may include an implementation of a networking environment. Program modules **42** generally carry out the functions and/or methodologies of embodiments of the invention as described herein.

[0176] Computer system/server **12** may also communicate with one or more external devices **14** such as a keyboard, a pointing device, a display **24**, etc.; one or more devices that enable a user to interact with computer system/server **12**; and/or any devices (e.g., network card, modem, etc.) that enable computer system/server **12** to communicate with one or more other computing devices. Such communication can occur via Input/Output (I/O) interfaces **22**. Still yet, computer system/server **12** can communicate with one or more networks such as a local area network (LAN), a general wide area network (WAN), and/or a public network (e.g., the Internet) via network adapter **20**. As depicted, network adapter **20** communicates with the other components of computer system/server **12** via bus **18**. It should be understood that although not shown, other hardware and/or software components could be used in conjunction with computer system/server **12**. Examples, include, but are not limited to: microcode, device drivers, redundant processing units, external disk drive arrays, RAID systems, tape drives, and data archival storage systems, etc.

[0177] The present invention may be a system, a method, and/or a computer program product. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the present invention.

[0178] The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device such as punch-cards or raised structures in a groove having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

[0179] Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

[0180] Computer readable program instructions for carrying out operations of the present invention may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk, C++ or the like, and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The computer readable program instructions may execute ‘entirely on’ the user's computer, ‘partly on’ the user's computer, as a stand-alone software package, ‘partly on’ the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be ‘connected to’ the user's computer

through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the present invention.

[0181] Aspects of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer readable program instructions.

[0182] These computer readable program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

[0183] The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

#### Multi-Focus Image Fusion With Background Removal

[0184] Described herein are methods for and systems for performing multi-focus image fusion with background removal. In particular, multi-focus image fusion generates a single best-focused image from one or more z-stacks of images (e.g., a plurality of z-stacks of images where each z-stack represents a different FOV of a plurality of FOVs). In some embodiments, a best in-focus image is generated for each color channel (e.g., red, yellow, green, blue, nUV) of an imaging instrument (e.g., an optofluidic instrument). Due to variations in the thickness, composition, and/or surface contours of a sample, the most in-focus z-slice (which can be represented as a z-slice index, e.g., an integer) of a z-stack of images taken of the sample may differ across FOVs of the sample (e.g., two adjacent FOVs have most in-focus z-slices that differ by one, two, three, four, or five z-slice indices). In some embodiments, the most in-focus z-slice may differ across a single FOV of the sample (e.g., if the FOV is divided into patches or sub-portions, two adjacent patches or sub-portions within the FOV may differ by one, two, three, four, or five z-slice indices). Because of this variation in focus across a sample, selecting a single z-slice to represent a FOV of the sample (or a plurality of FOVs of the sample) may result in diminished performance of downstream image processing where the most in-focus images (e.g., images with high contrast and well-defined edges) will produce better results, e.g., for nucleus and/or cell segmentation tasks. Moreover, images of biological samples (e.g., tissue samples, hydrogels having analytes immobilized therein, etc.) may include background signal (e.g., autofluorescence, reflected excitation light, etc.) that can be removed to further improve performance of downstream image processing. The systems and methods described herein advantageously divide each FOV into smaller patches, determine the best

in-focus z-slice index for each individual patch, and fuse the most in-focus patches together into a single most in-focus image that has high focus (e.g., high focus scores) across the entirety of the image. Additionally, or optionally, the systems and methods described herein advantageously remove background signal from the single most in-focus image to produce a high quality, in-focus, and background-removed image of a sample that can be used in downstream image processing, such as nucleus and cell segmentation tasks or to present to a user in a graphical user interface (e.g., a display). In some embodiments, the single most in-focus and background-removed image is generated for each color channel in an imaging instrument.

[0185] FIG. 6 illustrates a z-slice image **600** exhibiting distortion at the edges of the FOV due to field curvature. Field curvature (also called Petzval field curvature) is an optical aberration where light from a flat object is focused onto a curved surface instead of a flat plane, meaning that the sharpest image is formed on a curved focal plane, causing the edges of an image to appear blurry while the center remains sharp. Field curvature may be particularly noticeable in wide-angle lenses. As shown in FIG. 6, the z-slice image **600** includes a plurality of cells scattered across the FOV. Due to the field curvature of the objective lens used in the optofluidic instrument, regions at the edges of the FOV become defocused (because the focal plane is not perfectly flat across the FOV). In various embodiments, the field curvature can cause up to 3 z-slices of shift to the focal plane at various portions (e.g., the edges) of the FOV (assuming a 0.75  $\mu\text{m}$  step size between z-slices), thereby causing the cells at those portions of the FOV to appear out-of-focus. In one example of simple field curvature, a lens focuses rays of light from a curved field that passes through at least image plane A and image plane B. Thus, some portions of the resulting image will be most in-focus at image plane A and some portions of the resulting image will be most in-focus at image plane B.

[0186] FIG. 7 illustrates various cycles of an optofluidic instrument used to generate a final stain image using multi-focus image fusion. In various embodiments, the optofluidic instrument has a total of 17 cycles, which includes 15 decoding cycles, a blank cycle to image background, and a cell segmentation cycle to image various stains (e.g., antibody stains, cytoplasmic RNA stains and/or nuclear stains) needed for downstream cell segmentation tasks. Decoding cycles may also be referred to as probing cycles. In various embodiments, the optofluidic instrument has over 30 decoding cycles for detecting analytes (e.g., for an analysis of a 5000 gene panel). In various embodiments, the optofluidic instrument performs over 100 decoding cycles for detecting analytes (e.g., for a whole transcriptome analysis). In various embodiments, a first cycle **702** (e.g., a RNA decoding cycle) includes a nuclear stain (e.g., DAPI) cycle imaged in a nuclear stain color channel (e.g., a nUV channel), and four color channels for imaging RCPs (e.g., red, yellow, green, and blue channels). In various embodiments, additional cycles **704** (e.g., cycles **2** through **15**) are imaged in four color channels (e.g., red, yellow, green, and blue channels) for imaging analytes (e.g., RCPs associated with a RNA transcript). In various embodiments, a blank cycle **706** is performed after RNA decoding cycles have been performed (e.g., after all RNA decoding cycles have been performed). In various embodiments, the blank cycle **706** is performed after a predetermined number of decoding cycles has been performed such that the background signal is determined to be stable. For example, the predetermined number of decoding cycles may be 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15 cycles. In various embodiments, the predetermined number of decoding cycles is at least 8 cycles. In various embodiments, the blank cycle **706** is imaged in the nuclear stain color channel (e.g., nUV channel) and one or more other color channels to determine nuclear stain signal, autofluorescence signal, and/or background light (e.g., excitation light reflected at the sample and received at the image sensor through the emission filter). For example, the one or more channels includes the red channel, green channel, and blue channel (no yellow channel). One skilled in the art will recognize that any suitable number of color channels can be imaged to obtain suitable estimates of the nuclear stain signal, autofluorescence signal, and/or background light. In another example, the one or more channels includes the red channel, yellow channel, green channel, and blue channel. In various embodiments, a cell segmentation cycle **708** is performed. In various

embodiments, the cell segmentation cycle **708** is performed as the last cycle (i.e., after all RNA decoding cycles and the blank cycle). In various embodiments, the cell segmentation cycle **708** is imaged in the nuclear stain channel (e.g., nUV channel) and one or more other channels to thereby image one or more stains (e.g., membrane stain, cytoplasmic RNA stain, cytoplasmic protein stain, and/or nuclear stain) for downstream cell segmentation tasks. Exemplary methods of image segmentation and multi-modal cell segmentation can be found in U.S. patent application publication no. 2025-0061732 and U.S. patent application Ser. No. 19/037,114, each of which is incorporated by reference herein in its entirety. For example, the one or more channels includes the red channel, green channel, and blue channel (no yellow channel). In another example, the one or more channels includes the red channel, yellow channel, green channel, and blue channel. In various embodiments, when imaging proteins, the blank cycle and the cell segmentation cycle include DAPI imaging (e.g., in the nUV channel) and all color channels (e.g., in red, yellow, green, and blue channels). In various embodiments, the cell segmentation cycle **708** may be performed before the blank cycle **706**. In various embodiments, a final stain image **710** is generated using the multi-focus image fusion techniques described below.

[0187] In various embodiments, a first z-stack of images of a biological sample is received. In various embodiments, the first z-stack of images corresponds to a field of view (FOV). In various embodiments, the first FOV comprises a plurality of (i.e., two or more) patches. In various embodiments, the z-stack first images is based on at least one probing cycle of a biological sample in an imaging instrument. In various embodiments, the biological sample includes at least one cellular structure (e.g., one or more nuclei). In various embodiments, the first z-stack of images includes the cellular structure (e.g., the one or more nucleus).

[0188] In various embodiments, a first z-stack of images of a biological sample is acquired or received from an imaging instrument or other source. In various embodiments, the first z-stack of images includes background images of the biological sample, which may or may not include a background stain (e.g., DAPI). In various embodiments, the nuclear stain is applied and background signal is obtained in non-nuclear stain color channels (i.e., non-nUV color channels) to determine background signal that results from the nuclear stain. In various embodiments, the first z-stack of images is of a biological sample that lacks cytoplasmic, membrane, and/or nuclear staining. Preferably, the first z-stack of images is of a biological sample that lacks cytoplasmic and membrane staining. In various embodiments, the first z-stack of images includes images of a biological sample stained with nuclear staining, such as DAPI staining. In various embodiments, the background stain is only DAPI staining (i.e., no other stains emit light in the background stain image). In various embodiments, this first z-stack of images is referred to as a z-stack of background images.

[0189] In various embodiments, the first z-stack of images is captured after a predetermined number of probing cycles of an imaging instrument (where emitted light from fluorescently labelled rolling circle products is imaged). For example, the predetermined number of probing cycles may be 15 probing cycles. In various embodiments, the background signal of the sample may change during sample analysis. For example, the background signal after the first probing cycle may be different than the background cycle after the tenth probing cycle, but the background signal may not change (i.e., stabilizes) after a set number of cycles are completed. In various embodiments, the first z-stack of images representing the background is captured after the background signal of the sample is stable. In various embodiments, when the background signal (e.g., autofluorescence) becomes stable is empirically determined. For example, imaging can be performed of the biological sample with no staining to measured autofluorescence over 20 cycles and average intensity change per cycle can be determined. In various embodiments, the average intensity change decreases over time, and the sample has a stable background after ~8 cycles of imaging. In various embodiments, only one color channel (e.g., near UV) is imaged for the background z-stack of images. In various embodiments, multiple channels are imaged for the



background z-stack of images (e.g., nUV, red, green, and blue).

[0190] In various embodiments, a ZCYX (or ZCXY) imaging procedure is followed. Specifically, a z-stack of images is obtained at a fixed FOV (with fixed x- and y-values) for each color channel of a plurality of color channels. Then the fixed x- and/or y-values are changed to obtain images at a new FOV, repeating the z-stack imaging for each color channel, and this process continues for the entire tissue sample (or for a selected subset of FOVs). One benefit of this technique is that no image registration may be needed. Also, moving first in the z-direction, without changing the x- or y-values, can help minimize error since moving in the x- and y-directions can produce more error than moving in the z-direction. This imaging approach may be used to avoid computational image registration between channels, as the objective remains in the same XY position relative to the stage during capture of the z-stack across multiple color channels.

[0191] In various embodiments, a ZYXC (or ZXYC) imaging procedure is followed. Specifically, a z-stack of images is obtained at a fixed FOV (with fixed x- and y-values) for a single color channel before changing the x- and/or y-values to obtain images at a new FOV in the same color channel. After all FOVs are imaged for the entire tissue sample (or for a selected subset of FOVs) in the same color channel, the color channel is switched to the next color channel and a z-stack of images is obtained at all FOVs in the next color channel. One benefit of this method may be that the reduced number of switches between color channels results in less time elapsed for the entire procedure. Switching from one color channel to another color channel can take longer than moving in the z-direction through the sample. Also, moving first in the z-direction, without changing the x- or y-values, can help minimize error since moving in the x- and y-directions can produce more error than moving in the z-direction. Image registration may be required for FOVs across all color channels due to thermal drift in the sample and/or mechanical drift in the motion systems (e.g., z-stage, xy stage, etc.).

[0192] In some embodiments, the first imaging cycle is imaged using a ZCYX imaging procedure while subsequent imaging cycles for detection of analytes are imaged using a ZYXC imaging procedure. In some embodiments, imaging cycles for cell segmentation are imaged using a ZCYX imaging procedure. In some embodiments, imaging cycles for cell segmentation are imaged using a ZYXC imaging procedure. In some embodiments, a DAPI color channel (e.g., nUV) is imaged during (e.g., at the beginning of) each imaging cycle. In some embodiments, a DAPI color channel is imaged during the first imaging cycle for detection of analytes (but not subsequent imaging cycles for detection of analytes) and also during cell segmentation imaging cycles after the analyte detection imaging cycles are completed.

[0193] In various embodiments, the biological sample has no other staining, other than nuclear (e.g., DAPI) staining, during the blank cycle where a background z-stack of images (e.g., a multichannel z-stack of images) is obtained. For example, cytoplasmic staining (e.g., proteins or 18S ribosomal RNA) and/or membrane staining may not be applied to the sample and, thus, cytoplasmic and/or membrane staining is not imaged. Acquisition of multiple images, such as in the first z-stack of images representing the background signal of the biological sample, may be performed because the background of the images may change over cycles (but may stabilize after a predetermined number of cycles have been performed). In various embodiments, a base level of background signal is determined from one blank cycle and the background signal may be determined again from an additional, subsequent blank cycle to determine the extent to which the background signal changes over time (e.g., over one or more cycles of staining and imaging for cell segmentation).

[0194] In various embodiments, after the first z-stack of images is received or captured, a number of well-focused z-slices of the z-stack are determined. FIG. 8 depicts a visualization of a z-slice of a multi-channel z-stack of images of a biological sample (e.g., a background image with only DAPI staining). In particular, z-slice 30 (slice located at 22.5  $\mu\text{m}$  above the substrate surface) is being shown for a first background cycle. In various embodiments, distortion correction is performed on

the number of well-focused z-slices of the z-stack. For example, field curvature aberration correction is computationally performed. In another example, other optical aberrations can be computationally corrected for, such as, spherical aberration, coma, astigmatism, distortion, chromatic aberration. In various embodiments, deconvolution may be performed on the image to sharpen the image based on a point spread function that can be determined empirically for a particular optical system. In various embodiments, the first z-stack of images comprises multi-channel images (e.g., a z-stack of images in one or more color channels, such as, nUV, red, yellow, green, and/or blue channels). In various embodiments, Scale-Invariant Feature Transform (SIFT) features are extracted from these z-slices. In various embodiments, z-slices of the first z-stack of images are scored according to a focus score, such as a Vollath's F4 or Tenengrad. In various embodiments, a predetermined number (e.g., 3, 4, 5, 6, 7, 8, 9, 10, 11, etc.) of the highest scoring z-slices are determined. For example, the predetermined number may be three or more z-slices. In various embodiments, distortion correction may be performed based on information, such as distortion parameters, from the imaging instrument. For example, distortion correction may be performed to correct for distortion such as pincushion distortion as well as other types of distortion. In various embodiments, SIFT features from the determined/selected z-slices are then extracted. In various embodiments, the SIFT features are associated with features, textures, or points of at least one cellular structure shown in the z-slices, such as a morphology of the nuclei of one or more cells shown in the z-slices.

[0195] In various embodiments, a second z-stack of images of the biological sample is received. In various embodiments, the second z-stack corresponds to a second FOV, which may be the same FOV as the FOV of the first z-stack. In various embodiments, the second FOV includes a plurality of (i.e., two or more) patches, e.g., the same number of patches as the first FOV. In various embodiments, the second z-stack of images is based on at least one probing cycle of the biological sample in an imaging instrument. In various embodiments, the biological sample includes at least one cellular structure (e.g., one or more nucleus). In various embodiments, the second z-stack of images includes the cellular structure (e.g., the one or more nucleus), and this may be the same cellular structure in the first z-stack of images.

[0196] In various embodiments, the second z-stack of images of a biological sample is acquired or received from an imaging instrument or other source. In various embodiments, the second z-stack of images comprises multi-channel images (e.g., a z-stack of images in one or more color channels, such as, nUV, red, yellow, green, and/or blue channels). In various embodiments, the second z-stack of images includes images of the biological sample stained with at least one stain such that the at least one stain is illuminated in at least one illumination channel. In various embodiments, the at least one stain includes one or more fluorescent stains. The fluorescent stains may include one or more nuclear stain, such as DAPI, and/or one or more cytoplasmic stain, such as one or more ribosomal RNA stain (e.g., 18S), one or more lectin stain, one or more antibody stain, and/or the like. In various embodiments, this second z-stack of images is referred to as a z-stack of stained images.

[0197] In various embodiments, the second z-stack of images is captured after a predetermined number of probing cycles of an imaging instrument (where emitted light from fluorescently labelled rolling circle products is imaged). In various embodiments, the images are captured using a ZCYX (or ZCXY) imaging order. In particular, an imaging order of ZCYX (or ZCXY) images a z-stack of a single FOV in all color channels before moving in X and/or Y to the next FOV to image the next FOV in all color channels, and so on and so forth until all FOVs are imaged. This imaging approach may be used to avoid computational image registration between channels, as the objective remains in the same XY position relative to the stage during capture of the z-stack across multiple channels.

[0198] In various embodiments, after the second z-stack of images is received or captured, a number of well-focused stained z-slices of the z-stack are determined. FIG. 9 depicts a

visualization of a z-slice of a multi-channel z-stack of images of a stained biological sample. In particular, z-slice 30 (slice located at 22.5  $\mu\text{m}$  above the substrate surface) is being shown for a first cell segmentation cycle where one or more stains are imaged for downstream cell segmentation purposes (e.g., one or more cell membrane stain, one or more cytoplasmic stain, one or more nuclear stain, etc.). In various embodiments, distortion correction is performed on the number of well-focused z-slices of the z-stack. For example, field curvature aberration correction is computationally performed. In another example, other optical aberrations can be computationally corrected for, such as, spherical aberration, coma, astigmatism, distortion, chromatic aberration. In various embodiments, deconvolution may be performed on the image to sharpen the image based on a point spread function that can be determined empirically for a particular optical system. In various embodiments, the first z-stack of images comprises multi-channel images (e.g., a z-stack of images in one or more color channels, such as, nUV, red, yellow, green, and/or blue channels). In various embodiments, SIFT features are extracted from these stained z-slices. In various embodiments, z-slices of the z-stack of images are scored according to a focus score, such as a Vollath's F4 or Tenengrad. In various embodiments, a predetermined number (e.g., 3, 4, 5, 6, 7, 8, 9, 10, 11, etc.) of the highest scoring z-slices are determined. For example, the predetermined number may be three or more z-slices. In various embodiments, distortion correction is performed based on information, such as distortion parameters, from the imaging instrument. For example, distortion correction may be performed to correct for distortion such as pincushion distortion as well as other types of distortion. In various embodiments, SIFT features from the determined/selected stained z-slices are then extracted. In various embodiments, the SIFT features are associated with features or points of at least one cellular structure shown in the z-slices, such as a morphology of the nuclei of one or more cells shown in the z-slices, which may include the same or similar SIFT features associated with the first z-stack of images.

[0199] In various embodiments, the first z-stack of images and the second z-stack of images are registered to each other based on at least one cellular structure (e.g., at least one nucleus, at least one cell membrane, etc.) within the biological sample. In various embodiments, XYZ registration of the first z-stack of images and the second Z-stack of images is performed only in the nUV (DAPI) channel assuming that other channels are aligned with the nUV channel due to the ZCYX imaging order. In various embodiments, where ZYXC imaging is performed, registration is performed between the different channels imaged in the same cycle. At least one of the SIFT features associated with the cellular structure from the z-slices from the first z-stack, as described above, are matched with the SIFT features from the stained z-slices from the second z-stack, as described above. Based on this matching, an XYZ translation between the first z-stack and the second z-stack is determined using an image registration algorithm, such as the random sample consensus (RANSAC) algorithm. For example, the matching may produce a registered coordinate space, which includes an estimate of a translation of points in space between the first and the second z-stacks. In various embodiments, an image registration algorithm such as RANSAC produces a registered coordinate space for the first and the second z-stacks. For example, in various embodiments, the nucleus in the first z-stack of images may be registered to the nucleus in the second z-stack of images.

[0200] In various embodiments, a focus map is determined based on the second z-stack of images. In various embodiments, a focus map is determined for each color channel. For example, a focus map may be determined for a nUV channel, a blue channel, a green channel, a yellow channel, and/or a red channel. In various embodiments, the focus map indicates, for each of a plurality of patches of the selected FOV (which may be the first FOV, for example), one of the images (i.e., z-slices) of the second z-stack bringing into focus that patch of the FOV. In various embodiments, each patch is a block or group of a predetermined number of pixels (e.g., 8 by 8 pixel block, 16 by 16 pixel block, 32 by 32 pixel block, or 64 by 64 pixel block). In various embodiments, all patches will be square (have equal number of pixels for each side). In various embodiments, all patches will

have the same size of block (e.g., all patches will be 64 by 64 pixels).

[0201] In various embodiments, a multi-focus image fusion algorithm is applied to the second z-stack of images (e.g., using the selected subset of slices from the second z-stack of images). In various embodiments, the multi-focus image fusion algorithm is applied for each FOV. In various embodiments, the multi-focus image fusion algorithm is applied for each color channel. For example, with four color channels of red, yellow, green, and blue, the multi-focus image fusion algorithm may generate four focus maps for each FOV that is imaged. At a high-level, a multi-focus image fusion algorithm is an algorithm to computationally synthesize an all-in-focus image and/or a fused image from images taken at different focal planes (i.e., a z-stack of images). In particular, the resulting fused, most in-focus image includes a focused image synthesized from images taken with different focuses (i.e., images taken at different z-slices at a FOV). Multi-focus image fusion algorithms may have applications to volumetric imaging and, in particular, to microscopy applications where three-dimensional volumes are imaged (e.g., in situ analysis of three-dimensional samples). Any such multi-focus image fusion algorithm may have variations (e.g., based on the sample type, analytes being detected/quantified, etc.). The application of the multi-focus image fusion algorithm may be used to determine a focus map and generate an all-in-focus image and/or a fused image.

[0202] In various embodiments, a multi-focus image fusion algorithm is applied to the second z-stack of images using the following process.

[0203] As one step, in various embodiments, the most focused z-slice (i.e., image plane) of z-slices in the second z-stack of images is determined. For example, the most focused z-slice out of all (e.g., 40) z-slices in the second z-stack of images may be determined by scoring each z-slice according to a focus score, such as Vollath's F4 or Tenengrad, and the highest scoring z-slice is determined as the most focused z-slice. FIG. 10A shows a grayscale image of a z-slice having a z-index of 10 (i.e. the 11th z-slice in the z-stack from a reference slice having an index of 0, such as the glass slide). Where 20 adjacent z-slices are selected, the z-index of 10 may represent the most in-focus z-slice within the z-stack and 10 slices below the 11th z-slice and 10 slices above the 11th z-slice are selected for further processing, as explained in more detail below. In various embodiments, the Vollath's F4 focus score for an image is a single, unitless number representing how in-focus the image is across the entire image. The equation for Vollath's F4 is below:

$$[00001] F_{\text{Voll4}} = \frac{M-1}{i=1} \cdot \frac{N}{j=1} g(i,j)g(i+1,j) - \frac{M-2}{i=1} \cdot \frac{N}{j=1} g(i,j)g(i+2,j)$$

where  $g(i, j)$  is the gray level intensity of I pixel (i, j) in an image of size  $M \times N$ .

[0204] In various embodiments, because a focused image has high contents of higher frequencies, the focus score may be based on a measure of high frequencies (e.g., edges) within the image. In various embodiments, a Tenenbaum Gradient (Tenengrad) focus score is determined for a plurality of images (e.g., image slices) within a z-stack. The Tenengrad is a convolution of an image with vertical ( $S_{\text{sub.x}}$ ) and horizontal ( $S_{\text{sub.y}}$ ) Sobel operators. In various embodiments, the Tenengrad sums the square of all the magnitudes greater than a predetermined threshold. In various embodiments, all pixels are included in the summation. To determine a global measure over the whole image, the square of the gradient vector components are summed, as follows:

$$[00002] F_{\text{Tenengrad}} = S_x(x, y)^2 + S_y(x, y)^2$$

In various embodiments, a normalization factor is applied to the pixel values while determining the Tenengrad. In various embodiments, the normalization factor includes dividing each pixel value by a square root of the sum of the squared pixel values. In various embodiments, the normalization factor includes an intensity of the pixels (e.g., mean intensity, standard deviation, maximum intensity, minimum intensity, etc.)

[0205] In various embodiments, other suitable focus scores may be determined. In various embodiments, an entropy-based focus score may be determined for one or more (e.g., all) images within the z-stacks.

$$[00003] F_{\text{Entropy}} = - \sum_{\text{intensities}} \text{Math. } p(i) \log p(i)$$

[0206] In various embodiments, a Sum of Modified Laplace (SML) focus score may be determined for one or more (e.g., all) images within the z-stacks. An example of a SML is as follows:

$$[00004] F_{\text{SML}} = \text{Math. } \sum_x \text{Math. } L_x(x, y) \text{Math. } + \text{Math. } \sum_y L_y(x, y) \text{Math.}$$

[0207] In various embodiments, a focus score is based on the contrast of a image as the absolute difference of a pixel with its eight neighbors, summed over all the pixels of the image.

$$F_{\text{sub.contrast}} = \sum \sum C(x, y)$$

where the contrast  $C(x, y)$  for each pixel in the gray image  $I(x, y)$  is determined as:

$$[00005] C(x, y) = \text{Math. } \sum_{i=x-1}^{x+1} \sum_{j=y-1}^{y+1} \text{Math. } I(x, y) - I(i, j) \text{Math.}$$

[0208] In various embodiments, a focus score is based on the coefficients of the discrete cosine transform obtained after dividing the image into  $8 \times 8$  non overlapped windows and then averaging over all the  $8 \times 8$  Windows.

$$[00006] F_{M_e} = \frac{\text{Math. } M'_{B_e}}{\text{num. of } 8 \times 8 \text{ windows}}$$

where  $M'_{\text{sub.B.sub.e}}$  is determined from the DCT coefficients  $F(\omega, v)$  as:

$$[00007] M'_{B_e} = 1 - \frac{\text{Math. } F(\omega, v) \text{Math. }^2}{(\text{Math. } F(\omega, v))^2}$$

[0209] As another step, in various embodiments, a predetermined number of z-slices in the second z-stack adjacent to the most focused plane are selected. For example, 10 z-slices above and 10 z-slices below a most in-focus z-slice may be selected, for a total of 21 slices in the predetermined number of slices. In various embodiments, the predetermined number of z-slices together with the most in-focus plane is referred to as a selected set of adjacent images. In various embodiments, the selected set of adjacent images has a predetermined size, such as a size between 2 images and 41 images. For example, a predetermined number of z-slices above and below (e.g., +10) the most focused plane/z-slice may be determined and extracted to form the selected set of adjacent images. In various embodiments, the upper bound on the predetermined size depends on the tissue thickness. For example, thicker tissue samples may have a larger predetermined size of the selected set of images. Each of these images in the selected set of adjacent images may include associated corresponding z-index indicating its position within the second z-stack. In various embodiments, the z-index ranges from 0 to a maximum number of adjacent images (e.g., 0 to 20). Alternatively, the z-index ranges from  $-\text{max adjacent images}/2$  to a  $+\text{max adjacent images}/2$  (e.g., -10 to +10) where the zeroth index is the most in-focus z-slice. As explained above, this process of determining a most in-focus z-slice based on a focus score and then selecting a predetermined number of adjacent slices can be performed for each color channel of a plurality of color channels (e.g., nUV, blue, green, yellow, and/or red).

[0210] As another step, optionally, a denoising filter, such as a Gaussian filter, may be applied to each image of the selected set of adjacent images determined at the previous step. For example, the filter may be applied to denoise each z-slice image in the set.

[0211] As another step, in various embodiments, each of the selected set of adjacent images is divided into a number of image patches (2-D x-y dimension patches). In various embodiments, a most focused z-slice (which may be represented as a z-index, which is an incrementing integer assigned to each z-slice in the z-stack) is selected for each image patch. As used herein, a “patch” is a group of pixels, such as a square of  $64 \times 64$  pixels, in an image. In various embodiments, each of the images corresponding to the selected set of adjacent images is divided into the same plurality of patches. Each patch may be a portion of an image (i.e., a z-slice of a z-stack taken at a FOV) in the x-y plane that also corresponds to a z-index (varying in the z direction) of the selected set of adjacent images. In various embodiments, the z-index for each patch corresponds to a z-index of a z-slice within the second z-stack. In various embodiments, each patch may have the same shape,

such as a square shape, and may be about  $4 \times 4$  to  $128 \times 128$  pixels in size. For example, the number of image patches may include  $16 \times 16$  pixel denoised image patches,  $32 \times 32$  pixel denoised image patches,  $64 \times 64$  pixel denoised image patches, or  $96 \times 96$  pixel denoised image patches. In various embodiments,  $64 \times 64$  pixel patches was selected because this patch size captures the texture of a single cell (a single nucleus is about  $20 \mu\text{m} \times 20 \mu\text{m}$ , which is about 100 pixels by 100 pixels)

[0212] In various embodiments, the focus map is generated having the same size and number of patches as the selected set of adjacent images. For each patch in the focus map, a most-focused patch is selected from the patches in the same position across the selected set of adjacent images and a z-index of the most-focused patch is assigned to the respective patch in the focus map. Each patch in each image of the selected set of adjacent images (with varying z-indices) is scored according to a focus score, such as Tenengrad or a Vollath's F4, and the z-index of the highest scoring patch is determined. Performing such a technique for all patches, thereby assigning z-indices from the most in-focus patches from the selected set of images to a map, generates a raw focus index map, as shown in FIG. 10B. In various embodiments, the raw focus index map indicates the index of the z-slice associated with the selected image for each patch.

[0213] As another step, in various embodiments, outliers in the raw focus index map are removed to generate an outlier-removed focus index map. In various embodiments, for each patch (as a target patch) in the raw focus index map, z-indices of one or more neighboring patches are compared with the z-index of the target patch. That is, the z-index of a patch will be compared to the z-indices of neighboring patches, and this comparison will be performed for each patch in the raw focus map. In various embodiments, when the patch has more than a predetermined disparity (e.g., difference) between its associated z-index and a z-index associated with one or more neighboring patches the patch is determined to be a disparate patch. FIG. 10C illustrates a heat map of number of neighbors with an abrupt change. That is, for each patch, the color represents the number of neighbors (0, 1, 2, 3, or 4 neighbors) having z-indices that are above a predetermined disparity (e.g., greater than 5 z-index difference). In various embodiments, disparate patches are removed, as shown in FIG. 10D. In various embodiments, after removal, disparate patches are replaced with replacement values for the indices, as shown in FIG. 10E. In particular, the replacement values may be interpolated z-indices between neighboring patches (e.g., interpolated based on the z-indices of all 4 adjacent patches) such that the interpolated value is within the predetermined disparity. In various embodiments, the predetermined disparity is about 2 to about 10 z-indices. In various embodiments, the predetermined disparity is about 5 z-indices. In various embodiments, the predetermined disparity is about 10% to about 50% of the total number of well-focused stained z-slices selected. In various embodiments, the predetermined disparity is about 20% to about 30% of the total number of well-focused stained z-slices selected. In various embodiments, the predetermined disparity is about 25% of the total number of well-focused stained z-slices selected (e.g., predetermined disparity of 5 for a selection of 20 well-focused stained z-slices). In various embodiments, a z-index of a patch is removed and replaced when two or more (e.g., two) neighboring patches have z-indices with a predetermined disparity with the z-index of the patch. For example, if a patch has a z-index of 10, two neighboring patches with z-indices of 11 and two neighboring patches with z-indices of 16, then the patch having the z-index of 10 will be removed and replaced (e.g., replaced with an average of the pixel values, 13). In various embodiments, when the z-index of a patch is determined to disparate (after comparison with one or more z-indices of one or more neighboring patches), the patch is replaced by a z-index value that is based on a nearest neighbor z-index value. In various embodiments, if the z-index of the current patch is greater than a predetermined disparity from the z-index of a neighboring patch, and the number of such neighboring patches is greater than a threshold number of disparate neighbors, then the z-index of the current patch in the raw focus index map is replaced. For example, as shown in FIG. 10E and explained above, the replacement z-index value may be interpolated based on two or more neighboring patch z-index values. In various embodiments, the replacement z-index value is

an average of a number of (e.g., all four) neighboring patch z-index value(s). In various embodiments, the replacement z-index value is assigned one of the neighboring patch z-index values (rather than interpolating). After such outlier indices are removed (and replaced) from the raw focus index map, the resulting map is an outlier-removed focus index map.

[0214] As another step, in various embodiments, a filter (e.g., a median filter) is applied to the raw and/or outlier-removed focus index map(s) to produce a filtered focus index map. As shown in FIG. **10F**, a median filter is applied to the focus index map (after having outliers removed and replaced). In various embodiments, the filtered index map includes floating point values for the z-indices (rather than integers).

[0215] As another step, in various embodiments, the focus index map (e.g., the raw, outlier-removed or filtered focus index map) is upsampled to a full pixel resolution. As shown in FIG. **10G**, the median filtered focus map is upsampled to a full pixel resolution such that each pixel is assigned a z-index value (e.g., a floating-point value). For example, the raw, outlier-removed or filtered focus index map may be upsampled using linear interpolation. In particular, the per pixel value of the z-index may be interpolated based on the raw, outlier-removed or filtered focus index map. In various embodiments, the resulting focus index map, after upsampling the raw, outlier-removed or filtered focus index map, is referred to herein as an upsampled focus map generated by the application of a multi-focus image fusion algorithm. In various embodiments, the upsampled focus map is used to sample a z-stack of images. In various embodiments, the upsampled focus map includes floating point values. In various embodiments, a function, such as a round, floor, or ceiling function, is applied to the upsampled focus map before it is used to sample a z-stack, in order to change any floating-point values of the z-indices in the focus map to integer values of the z-indices. For example, a floor function can be applied to the upsampled focus map to sample the background fuse\_bg\_0, and then the background can be sampled again for fuse\_bg\_1 by adding one to each z-index in the focus map. In another example, a ceiling function can be applied to the upsampled focus map to sample the background fuse\_bg\_1, and then the background can be sampled again for fuse\_bg\_0 by subtracting one to each z-index in the focus map.

[0216] In various embodiments, the second z-stack of images is sampled using the upsampled focus map generated by the application of the multi-focus image fusion algorithm. Thus, the upsampled focus map is applied to the second z-stack of images. The result of applying the upsampled focus map to the z-stack of images may be referred to herein as a stained fused FOV image or an all-in-focus FOV image generated by applying the upsampled focus map generated by the multi-focus image fusion algorithm to the second z-stack of images. FIG. **10H** shows an example of a stained fused image (e.g., in a specific color channel) generated by applying the upsampled focus map to a plurality of z-stacks of images (representing a plurality of FOVs of a sample). As explained above, a plurality of upsampled focus index maps may be generated, where each upsampled focus index map represents a specific color channel. In summary, the multi-focus image fusion algorithm may assist in determining, for the second z-stack of images, for each pixel patch, which z-slice of the second z-stack of images to include in an all-in-focus/fused FOV image based on the upsampled focus map.

[0217] In various embodiments, the multi-focus image fusion algorithm is applied to the second z-stack of images using the following steps: [0218] 1. Find the most focused plane out of the 40 z-slices. (Focus score: Vollath's F4). [0219] 2. Extract  $\pm 10$  z-slices around the most focused plane. [0220] 3. Apply a Gaussian filter to each slice as a mild denoising step. [0221] 4. Make  $64 \times 64$  denoised image patches and find the most focused plane at each image patch. (Focus score: Tenengrad). This step creates a raw focus index map. [0222] 5. Find an abrupt change of focus index and replace them with nearest neighbors to generate an outlier-removed focus index map. [0223] 6. Apply a median filter to the outlier-removed focus index map to generate a filtered focus index map. [0224] 7. Upsample the filtered focus index map with linear interpolation to the full resolution to generate an upsampled focus index map. [0225] 8. Sample the original z-stack with

the upsampled focus index map.

[0226] In various embodiments, the first z-stack images is sampled with the upsampled focus map to generate an unstained fused image. In particular, the upsampled focus index map may be applied to the first z-stack of images. The result of this may be referred to herein as an unstained fused image generated by applying the upsampled focus index map generated by the multi-focus image fusion algorithm to the first z-stack of images.

[0227] Referring now to FIG. 11, in various embodiments, the first z-stack of images is sampled with the upsampled focus index map to generate a first unstained intermediate image 1104. In various embodiments, the first z-stack of images is sampled with a shifted version of the upsampled focus index map to determine a second unstained intermediate image 1106. In various embodiments, a convex combination of the first unstained intermediate image 1104 and the second unstained intermediate image 1106 may be determined to generate an unstained fused image 1107. In various embodiments, the unstained fused image 1107 is subtracted from the stained fused image 1102 to produce a subtracted image 1108, such as a background-subtracted image.

[0228] In various embodiments, the upsampled focus map, generated by the application of a multi-focus image fusion algorithm, is used to sample the first z-stack/z-stack of unstained images to produce a first unstained intermediate image. In various embodiments, a shifted version of this upsampled focus index map is used to sample the first z-stack/z-stack of unstained images to produce a second unstained intermediate image. For example, the upsampled focus index map with each z-index shifted up or down by a z-index of 1, may be used to sample the first z-stack of images. In various embodiments, a function, such as a round, floor, or ceiling function, is applied to the upsampled focus index map before it is used to sample a z-stack or before the upsampled focus index map is shifted, in order to change any floating-point values in the upsampled focus index map to integer values.

[0229] In various embodiments, a convex combination of the first unstained intermediate image and the second unstained intermediate image is determined by sampling the first z-stack of images using the upsampled focus index map and a shifted version of the upsampled focus index map. In various embodiments, a convex combination of b and c is determined using the formula— $y(a) = (1-a)b + ac$ , where  $0 \leq a \leq 1$ . This formula may be applied to images, where b and c are images, each with multiple pixel points. In various embodiments, an algorithm, such as the random sample consensus (RANSAC) algorithm, is used to determine the convex combination of the first and second intermediate images. For example, multiple convex combinations of the first and the second intermediate images may be determined, using the algorithm, and each may be scored according to a focus metric or focus score. The highest scoring convex combination of the intermediate images may be determined as the best convex combination and used as the unstained fused image.

[0230] In various embodiments, the unstained fused image is subtracted from the stained fused image produced by applying the multi-focus image fusion algorithm. For example, pixel-by-pixel subtraction may be performed, whereby each pixel of the unstained fused image may be subtracted from each corresponding pixel of the stained fused image. A subtracted image, such as a background-subtracted image, may be produced as a result of this subtraction being performed.

[0231] In various embodiments, shifted versions of the upsampled focus map (for removing the background signal in the convex combination) can be determined as follows: [0232] 1. Based on the (floating-point) focus map, sample a slice with the (floating-point) focus map from the background z-stack with linear interpolation. Similarly, sample (+1) and (−1) z-slices from the focus map, resulting in three sampled slices in total. [0233] 2. From the above (+1) and below (−1) slices, compute the central difference as the estimate of the z-gradient image. In some embodiments, a forward or backward difference can be used as an alternative to central difference. [0234] 3. Determine the best-fitted background image with RANSAC by using the first order Taylor approximation model with the estimated z-gradient image.

In an example, where fuse\_bg\_1 is sampled with “focus\_index-0.5” and fuse\_bg\_2 is sampled with



“focus\_index+0.5” (without a floor operation), the result may be considered as the first order Taylor approximation model with the forward difference.

[0235] In various embodiments, each subtracted image of a plurality of subtracted images (representing all FOVs of a sample in a particular color channel) may be stitched together based on a FOV of that subtracted image to produce a finalized subtracted focused image. Additionally, or optionally, the resulting background-subtracted images for each color channel are combined with background-subtracted images from all other color channels to generate a multicolor, background-subtracted, multi-focus, fused image. FIG. 12 illustrated a resulting multicolor, background subtracted, multi-focus, fused image that can be provided to downstream image processing tasks (e.g., nucleus segmentation, cell membrane segmentation, etc.) and/or presented to a user via a graphical user interface (e.g., a display on the optofluidic instrument).

[0236] As discussed above, images may be captured by the imaging instrument using a ZCYX (or ZCXY) imaging order, in which the instrument images a z-stack of a single FOV in all color channels before moving in X or Y to the next FOV to image the next FOV in all color channels, and so on and so forth until all FOVs are imaged. For each captured FOV of the biological tissue sample, and for each channel, a subtracted image may be produced, as described herein, and it may be stitched together with other of the subtracted images produced for the other FOVs, to produce a finalized subtracted all-in-focus image.

[0237] FIG. 13 is a flowchart 1300 illustrating a method of image fusion, according to embodiments of the present disclosure. At 1302, a first z-stack of images of a biological sample may be received. The first z-stack of images may correspond to a first field of view. The first field of view may comprise a plurality of patches. At 1304, a second z-stack of images of the biological sample may be received. The second z-stack of images may correspond to the first field of view. At 1306, a focus map may be determined based on the second z-stack. The focus map may indicate, for each of a plurality of patches of the first field of view, one of the images of the second z-stack bringing into focus that patch of the first field of view. At 1308, the focus map may be applied to the first z-stack to generate a first fused image. At 1310, the focus map may be applied to the second z-stack to generate a second fused image. At 1312, the first fused image may be subtracted from the second fused image to produce a subtracted image.

[0238] In various embodiments, the first z-stack of images may comprise background images of the biological sample, the second z-stack of images may comprise images of the biological sample stained with at least one stain, and the subtracted image may comprise a background-subtracted image. In various embodiments, the background images of the biological sample that may include nuclear staining, such as a DAPI stain. The first z-stack of images may comprise images of the biological sample that lacks cytoplasmic, membrane, and/or nuclear staining. The first z-stack of images may comprise images of the biological sample stained with a nuclear stain. The nuclear stain may comprise DAPI. The first z-stack of images may comprise images of the biological sample stained solely with DAPI. The second z-stack of images may comprise images of the biological sample stained with one or more fluorescent stain. The one or more fluorescent stains may comprise a nuclear stain. The one or more fluorescent stains may comprise one or more cytoplasmic stain. The one or more cytoplasmic stain may comprise at least one of: one or more ribosomal RNA stain, one or more lectin stain, and one or more antibody stain. The first z-stack of images and second z-stack of images may be acquired based on at least one probing cycle of the biological sample in an imaging instrument. The biological sample may comprise at least one cellular structure. The first z-stack of images and the second z-stack of images may be registered to each other based on the at least one cellular structure. The at least one cellular structure may comprise a nucleus. The first z-stack of images and the second z-stack of images each may comprise the nucleus. The registering may comprise registering the nucleus of the first z-stack to the nucleus of the second z-stack. The registering may comprise a scale-invariant feature transform (SIFT). The registering may comprise applying random sampling consensus (RANSAC).

[0239] In various embodiments, determining the focus map may include: determining a first focus metric for each image of the second z-stack; selecting one image of the second z-stack having a highest value of the first focus metric; selecting a set of adjacent images of the second z-stack including the selected image; determining a second focus metric for each of the plurality of patches of each of the selected set of images; and for each patch of the plurality of patches, selecting one of the selected set of images having a highest value of the second focus metric for that patch. The first focus metric may comprise Vollath's F4. The first focus metric may comprise a Tenengrad. The second focus metric may comprise a Tenengrad. The second focus metric may comprise Vollath's F4. Each patch in the plurality of patches may comprise a same shape. The shape may comprise a square shape. The shape may be about 16×16 pixels to about 128×128 pixels in size. Each selected image of the selected set of images may have an associated index indicating a position within the second z-stack of images. The focus map may indicate the index associated with the selected image for each patch. One or more patch having more than a predetermined disparity between its associated index and the indices associated with two or more neighboring patches may be identified. The index associated with the identified one or more patch may be replaced with an interpolated index based on the indices of the two or more neighboring patches. The predetermined disparity may be 2, 3, 4, 5, 6, 7, 8, 9, or 10. A median filter may be applied to the indices of the focus map. The focus map may be upsampled to pixel resolution. The upsampling may comprises linear interpolation. The set of adjacent images may be of a predetermined size. The predetermined size may be 2 images to 31 images. The predetermined size may be 21 images. A denoising filter may be applied to each image of the selected set of images. Applying the focus map to the first z-stack of images may comprise: sampling the first z-stack of images according to the focus map to generate a first intermediate image; generating a shifted focus map from the focus map; sampling the first z-stack of images according to the shifted focus map to generate a second intermediate image; and determining a convex combination of the first intermediate image and the second intermediate image to generate the first fused image. Determining the convex combination may comprise a random sampling consensus. The subtracted image may be stitched together with one or more additional subtracted images based on the field of view and a color channel of the subtracted image.

[0240] FIG. 12 is a flowchart 1200 illustrating a method of image fusion, according to embodiments of the present disclosure. At 1202, a z-stack of images of a biological sample may be received. The z-stack of images may correspond to a field of view. At 1204, a focus map may be determined based on the z-stack of images, wherein determining the focus map may comprise: determining a focus metric for each image of the z-stack of images; selecting one image of the z-stack having a highest value of the focus metric; selecting a set of adjacent images of the z-stack including the selected image; dividing each image in the selected set of images into a plurality of patches; determining a second focus metric for each of the plurality of patches of each of the selected set of images; and for each patch of the plurality of patches, selecting one of the selected set of images having a highest value of the second focus metric for that patch. At 1206, the focus map may be applied to the z-stack to generate a fused image.

[0241] The z-stack of images may comprise images of the biological sample stained with one or more fluorescent stain. The one or more fluorescent stains may comprise a nuclear stain. The one or more fluorescent stains may comprise one or more cytoplasmic stain. The one or more cytoplasmic stain may comprise at least one of: one or more ribosomal RNA stain, one or more lectin stain, and one or more antibody stain. The first focus metric may comprise Vollath's F4. The second focus metric may comprise Tenengrad. The first focus metric may comprises Tenengrad. The second focus metric may comprise Vollath's F4. Each patch in the plurality of patches may comprise a same shape. The shape may comprise a square shape. The shape may be about 16×16 pixels to about 128×128 pixels in size. Each selected image of the set of selected images may have an associated index indicating a position within the z-stack of images. The focus map may indicate

the index associated with the selected image for each patch. One or more patch having more than a predetermined disparity between its associated index and the indices associated with two or more neighboring patches may be identified. The index associated with the identified one or more patch may be replaced with an interpolated index based on the indices of the two or more neighboring patches. The predetermined disparity may be 2, 3, 4, 5, 6, 7, 8, 9, or 10. A median filter may be applied to the indices of the focus map. The focus map may be upsampled to pixel resolution. The upsampling may comprise linear interpolation. The set of adjacent images may be of a predetermined size. The predetermined size may be 2 images to 31 images. The predetermined size may be 21 images. A denoising filter may be applied to each image of the selected set of images. [0242] The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

[0243] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

## Claims

1. A method of image fusion, the method comprising: receiving a first z-stack of images of a biological sample, the first z-stack of images corresponding to a first field of view, the first field of view comprising a plurality of patches; receiving a second z-stack of images of the biological sample, the second z-stack of images corresponding to the first field of view; determining a focus map based on the second z-stack, the focus map indicating, for each of a plurality of patches of the first field of view, one of the images of the second z-stack bringing into focus that patch of the first field of view; applying the focus map to the first z-stack to generate a first fused image; applying the focus map to the second z-stack to generate a second fused image; and subtracting the first fused image from the second fused image to produce a subtracted image.
2. The method of claim 1, wherein the first z-stack of images comprises background images of the biological sample, the second z-stack of images comprises images of the biological sample stained with at least one stain, and the subtracted image comprises a background-subtracted image.
3. (canceled)
4. The method of claim 1, wherein the first z-stack of images comprises images of the biological sample stained with a nuclear stain.
- 5-6. (canceled)
7. The method of claim 1, wherein the second z-stack of images comprises images of the biological sample stained with one or more fluorescent stain.

**8-10.** (canceled)

**11.** The method of claim 1, wherein the first z-stack of images and second z-stack of images are acquired based on at least one probing cycle of the biological sample in an imaging instrument.

**12.** The method of claim 1, wherein the biological sample comprises at least one cellular structure, the method further comprising: registering the first z-stack of images and the second z-stack of images to each other based on the at least one cellular structure.

**13.** The method of claim 12, wherein the at least one cellular structure comprises a nucleus, wherein the first z-stack of images and the second z-stack of images each comprise the nucleus, wherein registering comprises registering the nucleus of the first z-stack to the nucleus of the second z-stack.

**14-15.** (canceled)

**16.** The method of claim 1, wherein determining the focus map comprises: determining a first focus metric for each image of the second z-stack; selecting one image of the second z-stack having a highest value of the first focus metric; selecting a set of adjacent images of the second z-stack including the selected image; determining a second focus metric for each of the plurality of patches for each of the selected set of images; and for each patch of the plurality of patches, selecting one of the selected set of images having a highest value of the second focus metric for that patch.

**17.** The method of claim 16, wherein the first focus metric comprises Vollath's F4.

**18.** The method of claim 17, wherein the second focus metric comprises Tenengrad.

**19-22.** (canceled)

**23.** The method of claim 1, wherein each patch of the plurality of patches has a size of about  $16 \times 16$  pixels to about  $128 \times 128$  pixels.

**24.** The method of claim 16, wherein each selected image of the selected set of images has an associated index indicating a position within the second z-stack of images, and wherein the focus map indicates the index associated with the selected image for each patch.

**25.** The method of claim 24, further comprising: identifying one or more patch having more than a predetermined disparity between its associated index and the indices associated with two or more neighboring patches; and replacing the index associated with the identified one or more patch with an interpolated index based on the indices of the two or more neighboring patches.

**26.** The method of claim 25, wherein the predetermined disparity is 2, 3, 4, 5, 6, 7, 8, 9, or 10.

**27.** The method of claim 24, further comprising: applying a median filter to the indices of the focus map.

**28.** The method of claim 24, further comprising: upsampling the focus map to pixel resolution.

**29.** The method of claim 28, wherein said upsampling comprises linear interpolation.

**30.** The method of claim 1, wherein the set of adjacent images is of a predetermined size.

**31.** The method of claim 30, wherein the predetermined size is 2 images to 31 images.

**32.** (canceled)

**33.** The method of claim 1, further comprising: applying a denoising filter to each image of the selected set of images.

**34.** The method of claim 1, wherein applying the focus map to the first z-stack of images comprises: sampling the first z-stack of images according to the focus map to generate a first intermediate image; generating a shifted focus map from the focus map; sampling the first z-stack of images according to the shifted focus map to generate a second intermediate image; and determining a convex combination of the first intermediate image and the second intermediate image to generate the first fused image.

**35.** The method of claim 34, wherein determining the convex combination comprises a random sampling consensus.

**36.** The method of claim 1, further comprising stitching together the subtracted image with one or more additional subtracted images based on the field of view and a color channel of the subtracted image.

**37.** A computer program product comprising a computer readable storage medium having program instructions embodied therewith, the program instructions executable by a processor to cause the processor to perform a method comprising: receiving a first z-stack of images of a biological sample, the first z-stack of images corresponding to a first field of view, the first field of view comprising a plurality of patches; receiving a second z-stack of images of the biological sample, the second z-stack of images corresponding to the first field of view; determining a focus map based on the second z-stack, the focus map indicating, for each of a plurality of patches of the first field of view, one of the images of the second z-stack bringing into focus that patch of the first field of view; applying the focus map to the first z-stack to generate a first fused image; applying the focus map to the second z-stack to generate a second fused image; and subtracting the first fused image from the second fused image to produce a subtracted image.

**38.** A system comprising: an imaging instrument; and a computing node operatively coupled to the imaging instrument and comprising a computer readable storage medium having program instructions embodied therewith, the program instructions executable by a processor to cause the processor to perform a method comprising: receiving a first z-stack of images of a biological sample, the first z-stack of images corresponding to a first field of view, the first field of view comprising a plurality of patches; receiving a second z-stack of images of the biological sample, the second z-stack of images corresponding to the first field of view; determining a focus map based on the second z-stack, the focus map indicating, for each of a plurality of patches of the first field of view, one of the images of the second z-stack bringing into focus that patch of the first field of view; applying the focus map to the first z-stack to generate a first fused image; applying the focus map to the second z-stack to generate a second fused image; and subtracting the first fused image from the second fused image to produce a subtracted image.

**39-60.** (canceled)

---